CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

Department of Computer Science

Programme Structure & Syllabus
Bachelor of Computer Applications
(BCA/Honours/ Honours with Research)
Batch 2025-2029

CHRIST (Deemed to be University), Bangalore.
Karnataka, India
www.christuniversity.in

Department Overview

Department of Computer Science of CHRIST (Deemed to be University) strives to shape outstanding computer professionals with ethical and human values to reshape nation's destiny. The training imparted aims to prepare young minds for the challenging opportunities in the IT industry with a global awareness rooted in the Indian soil, nourished and supported by experts in the field.

Vision

The Department of Computer Science endeavors to imbibe the vision of the University "**Excellence and Service"**. The department is committed to this philosophy which pervades every aspect and functioning of the department.

Mission

"To develop IT professionals with ethical and human values". To accomplish our mission, the department encourages students to apply their acquired knowledge and skills towards professional achievements in their career. The department also moulds the students to be socially responsible and ethically sound.

Introduction to the Programme

Bachelor of Computer Applications (BCA) is an undergraduate program that focuses on building software professionals with strong practical and theoretical knowledge in Computer Science. The curriculum provides options for students to choose multiple electives depending on their interest of study. Interdisciplinary Courses on Mathematics, Statistics and Financial management enhance the breadth of domain knowledge.

The program is backed by a well-motivated set of teachers and an upgraded lab facility. Students get opportunities to be part of in-house projects/research projects/consultancy projects. The sixth semester project would be a capstone project with deliverables in the form of a project/article/patent.

Students with good academic records are encouraged to take up a fast track so that they can go for a fulltime industry internship during the sixth semester. Ample opportunities with credit transfers are available for students who would like to go on a student exchange to foreign universities for a short semester.

The exit options after the first year with a certificate on "Basic Programming Skills" and the second year with a diploma in "Computer Applications" and a BCA degree after the third year, the program offers a fourth year with a BCA Honours degree with Research as on option is designed for better employment as either as an application developer or developer with research capabilities.

Programme Objectives

This programme is conceptualised and designed based on the strong commitment of the department to provide better quality education to the students. The principal objectives of this course are to:

- Provide strong foundations in fundamentals of computer science and applications for employability and/or further graduation.
- Empower students with competencies in creative thinking and problem solving, interpersonal communication and managerial skills.
- Facilitate overall understanding of the technological development with legal and ethical issues.

- Equip the students in providing professional solutions to real-time problems.

Ethics and Human Values

1. Only proprietary or open source software would be used for academic teaching and learning purposes.
2. Copying of programs from internet, friends or from other sources is strictly discouraged since it impairs development of programming skills.
3. Unique Practical (Domain based) exercises ensures that the students don't involve in code plagiarism.
4. Projects undertaken by students during the course are done in teams to improve collaborative work and synergy between team members.
5. Projects involve modularization which initiates students to take individual responsibility for common goals.
6. Passion for excellence is promoted among the students, be it in software development or project documentation.
7. Giving due credit to sources during the seminar and research assignment is promoted among the students
8. The course and its design enforce the practice of good referencing technique to improve the sense of integrity.
9. Courses involving group discussions and debates on ethical practices and human values are designed to sensitize the students in dealing with customers and members within the organization.

Programme Outcomes:

PO1: Acquire and Apply Knowledge: Ability to understand and apply the fundamental principles, concepts and methods in key areas of Computer Applications and multidisciplinary fields.

PO2: Problem Analysis: Ability to analyze real-time problems using various tools and techniques.sdf

PO3: Design and Development: Ability to design and develop solutions to meet the desired needs.

PO4: State-of-art Technologies: Ability to adapt and apply emerging tools and technologies.

PO5: Entrepreneurship and Innovation: Ability to provide sustainable and innovative solutions for real-time problems.

PO6: Lifelong Learning: Ability to engage in continuous reflective learning in the context of technological advancement.

PO7: Communication and Team Building: Ability to demonstrate effective communication and interpersonal skills.

PO8: Ethics and Social Responsibility: Ability to integrate ethical and human values to become a socially responsible citizen.

Programme Eligibility

- Basic eligibility for the programme is a pass at the +2 level (Karnataka PUC / ISC / CBSE / NIOS / State Boards) from any recognised Board in India.
- Candidates writing the +2 examinations in March-May 2025 may apply with their class X and XI marks.

- Students pursuing International curriculum must note that eligibility is according to AIU stipulations:
- Applicants pursuing IB curriculum must have 3 HL and 3 SL with 24 credits.
- Applicants pursuing GCE / Edexcel must have a minimum of 3 A levels with a grade not less than C.

PROGRAMME STRUCTURE FOR BCA -BATCH 2025-29

SEMESTER - 1								
Course code	**Title of the Course**	**T+P**	**Course Type**	**Hrs/ Week**	**Marks**	**Credits**	**Total Hours**	**Course Category**
BCA001- 1	C Programming	2+0	BRIDGE COURSES	2	-		30	T&P
BCA101-1	Foundational Mathematics-1	3+0	Major Core	3	100	3	45	T
BCA102-1	Object Oriented Programming Using C++	3+2	Major Core	5	100	4	75	T&P
BCA203-1	Principles of Software Development and DBMS	3+2	Major Core	5	100	4	75	T&P
	Multidisciplinary Course	3+0	MDC	3	50	3	45	T
BCA181-1	Academic Writing	2+0	AEC	2	100	2	30	T
BCA161-1	Web Technologies	2+2	SEC	4	100	3	60	T&P
HED-NEP181-1	Holistic Education Development – I	1+0	VAC	1	-	1	15	T
EVS181-1	Environmental Studies	1+0	VAC	1	50	1	15	T
	TOTAL			**24**		**21**	**360**	

SEMESTER - 2								
Course code	**Title of the Course**	**T+P**	**Course Type**	**Hrs/ Week**	**Marks**	**Credits**	**Total Hours**	**Course Category**
BCA104-2	Discrete Mathematics	3+0	Major Core	3	100	3	45	T
BCA105-2	Operating System	4+0	Major Core	4	100	4	60	T
BCA105-2	Data Structures Using C++	3+2	Major Core	5	100	4	75	T&P
	Multidisciplinary Course	3+0	MDC	3	100	3	45	T
BCA181-2	Professional English	2+0	AEC	2	50	2	30	T
BCA262-2	Web Stack Development	2+2	SEC	4	100	3	60(30 theory +30 lab)	T&P
HED-NEP181-2	Holistic Education Development – II	1+0	VAC	1	-	1	15	T
UOI181-2	Understanding India	1+0	VAC	1	50	1	15	T
	TOTAL			**23**		**21**	**285**	

SEMESTER – 3

Course code	Title of the Course	T+P	Course Type	Hrs/ Week	Marks	Credits	Total Hours	Course Category
STA306-3	Statistics for Computer Applications	3+0	Major Core	3	100	3	45	T
Industry Integrated Course								
BCA207A-3	Robotic Process Automation							
BCA207C-3	Generative AI Tools and Creative Computing	1+2	Major Core	3	100	2	45	T&P
BCA208-3	Programming with Java	3+2	Major Core	5	100	4	75	T&P
BCA209-3	Python Programming	3+2	Major Core	5	100	4	75	T&P
BCA210-3	Computer Networks	4+0	Major Core	4	100	4	60	T
	Multidisciplinary Course (Swayam)	3+0	MDC			3		T
Elective-I(Language)								
FRE181-3	French							
GER181B-3	German - I	2+0	AEC	2	50	2	30	T
HED181-3	Holistic Education – III	1+0	VAC	1	50	1	15	T
BCA281-3	Summer Internship	Extra Credit	INTERNSHIP	4	100	4	60	P
	TOTAL			**27**		**23**		

SEMESTER – 4								
Course code	**Title of the Course**	**T+P**	**Course Type**	**Hrs/ Week**	**Marks**	**Credits**	**Total Hours**	**Course Category**
BBA220-4	Financial Accounting for Computer Application	3+0	Major Core	3	100	3	45	T
BCA211-4	Mobile Applications Development	3+2	Major Core	5	100	4	75	T&P
BCA363-4	Visual Programming	3+0	SEC	3	100	3	45	T
Elective II: (Choose any one course)								
BCA313A-4	Internet Of Things							
BCA313B-4	Augmented Reality And Virtual Reality							
BCA313C-4	Drone Programming	3+2	Core Elective	5	100	4	75	T&P
BCA381-4	DBMS Project	0+4	Major Core	4	100	2	60	P

BCA263-4	Service Learning		VAC	2	100	2	30	Theory–Community Engagement
Elective-III(Language)								
FRE181-4	French							
GER181B-4	German - II	2+0	AEC	2	50	2	30	T
HED181-4	Holistic Education – IV	1+0	VAC	1	50	1	15	T
	TOTAL			**25**		**21**	**375**	

SEMESTER – 5								
Course code	**Title of the Course**	**T+P**	**Course Type**	**Hrs/ Week**	**Marks**	**Credits**	**Total Hours**	**Course Category**
BCA314-5	Data Engineering	4+0	Major Core	4	100	4	60	T
BCA315-5	Applied Cloud Computing and Architecture	4+2	Major Core	6	100	5	90	T&P
BCA316-5	Digital Twin	3+2	Major Core	5	100	4	75	T&P
BCA317-5	Artificial Intelligence	4+0	Major Core	4	100	4	60	T
BCA481-5	Specialization Project	0+4	Project	4	50	2	60	P
Elective-IV (Choose any one course)								
BCA318A-5	Computer Vision							
BCA318B-5	Natural Language Processing							
BCA318C-5	Business Intelligence	4+2	Core Elective	6	100	5	90	T&P
	Total			**29**		**24**	**435**	

SEMESTER – 6								
Course code	**Title of the Course**	**T+P**	**Course Type**	**Hrs/Week**	**Marks**	**Credits**	**Total Hours**	**Course Category**
BCA320-6	Cyber Security	4+0	Major Core	4	100	4	60	T
BCA321-6	Design And Analysis Of Algorithm	4+0	Major Core	4	100	4	60	T
BCA484-6	Industry Internship		Internship		300	12		P
	Total			**8**		**20**		

SEMESTER – 7								
Course code	**Title of the Course**	**T+P**	**Course Type**	**Hrs/ Week**	**Marks**	**Credits**	**Total Hours**	**Course Category**
BCA401-7	Advanced Database Technologies	4+0	Major Core	4	100	4	60	T
BCA411-7	Research Methods in Machine Learning	3+2	Major Core	5	100	4	75	T&P
Elective1								
BCA402A-7	Blockchain Technology							
BCA402B-7	Digital Forensics							
BCA402C-7	Ethical Hacking							
BCA402D-7	Resource Management Techniques	4+0	Elective	4	100	4	60	T
Elective2								
BCA412A-7	Big Data Analytics							
BCA412B-7	Game Design and Development							
BCA412C-7	Computational Statistics							
BCA412D-7	Generative AI	3+2	Elective	5	100	4	75	T&P
BCA481H-7	Sustainable Computing Project		Project					
BCA482R-7	Dissertation I	0+12	Project	12	100	6	180	P
	Total			**30**		**22**		

SEMESTER – 8								
Course code	**Title of the Course**	**T+P**	**Course Type**	**Hrs/ Week**	**Marks**	**Credits**	**Total Hours**	**Course Category**
BCA403-8	Advanced Data Communication & Network	4+0	Major Core	4	100	4	60	T
BCA413-8	DevOps and Automation Framework	3+2	Major Core	5	100	4	75	T&P
Elective3								
BCA404A-8	High Performance Computing							
		4+0	Elective	4	100	4	60	T

BCA404B-8	Quantum Computing							
BCA404C-8	Cryptography and Network Security							
BCA404D-8	Edge Computing							
Elective4								
BCA414A-8	Big Data Analytics							
BCA414B-8	Drone Programming							
BCA414C-8	Pattern Recognition							
BCA414D-8	Agentic AI	3+2	Elective	5	100	4	75	T&P
BCA483H-8	Major Project		Project					
BCA484R-8	Dissertation II	0+12	Project	12	100	6	180	P
	Total			**30**		**22**		

SEMESTER 1

BCA001- 1 **C PROGRAMMING**

Teaching Hours: 30 hours

Course Description

This course introduces students to the C programming language, covering its history, features, data types, and program structure. Students will learn to apply decision control, loop structures, and various operators to create basic programs.

Course Objectives

By the end of this course, students will understand the history, evolution, and key features of the C programming language while identifying fundamental data types, operators, and program structures. They will develop basic programs using variables, constants, and input/output functions, implement decision control structures such as if-else and switch statements, and apply loop constructs like for, while, and do-while. Additionally, students will utilize various operators, analyze and debug programs for errors, demonstrate problem-solving skills through structured programming, and explore best practices for code readability and efficiency.

Course Outcome

CO1: Infer on the step for problem solving using programming language

CO2:Understand the fundamentals of C programming, including its history and features of C

CO3: Discuss and Demonstrate C programs using Tokens, variables and keywords and identifiers

CO4:Apply decision control statements, l and various operators to write basic C programs.

CO5: Interpret on loop control structures in C language

Prerequisite: Knowledge of computers

Unit 1

Introduction to Algorithm - flowchart -programming

Introduction to C

Types of Programming Language- History of C, Features of C, C Tokens, variables and keywords and identifiers

Unit 2

Data types & operators

Data Types, Type declaration, Different Operators in C - Arithmetic, Logical, Relational, Bitwise, Conditional, Expressions, Hierarchy of operations.

Unit 3

Control structures

Decision control statements- if, switch, conditional operator statement. Loop control structures- while, do-while, for loop, Break statement

Infosys springboard course:
https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138432370393743 3634517_shared/overview

Essential Reading
[1] **"Programming in ANSI C" by E. Balagurusamy**: The most recent edition is the **9th edition**, published in **2022** by McGraw Hill.

[2] **"Let Us C" by Yashavant Kanetkar**: The latest edition is the **19th edition**, published in **December 2022** by BPB Publications.

Evaluation Pattern:

CIA - 50%
ESE 50%

Course Name :	**Foundational Mathematics**	**Course Code:**	BCA101-1
Course Duration :	**45**	**Credits:**	**3**

Course Description: This course aims at introducing the students to the world of Discrete Mathematics. It includes topics like Mathematical Logic, Method of proofs, Mathematical induction, Permutations and combinations and Binomial coefficients. Also, this course emphasizes general techniques of problem-solving and explores the creation of mathematical patterns.

Course Objectives: This course will help the learner to

This course will help the learner to

COBJ1. Understand propositional logic, predicates, and quantifiers.
COBJ2. Apply proof techniques to mathematical problems.
COBJ3. Use mathematical induction for proving statements.
COBJ4. Solve problems using permutations, combinations, and binomial coefficients.

Course Outcomes: On successful completion of the course, the students should be able to

CO1. Reformulate statements into formal logic using propositional and predicate calculus and assess the validity of arguments.
CO2. Determine the validity of statements using propositional and predicate logic rules.
CO3. Construct elementary proofs using induction, including strong induction.
CO4. Apply counting principles, including the pigeonhole principle, permutations, and combinations, to solve problems.
CO5. Use logical structures and quantifiers to construct valid and clear mathematical arguments.

Pedagogy: Lectures, discussions, problem solving and seminars

Syllabus:

UNIT 1:	**Propositional Logic, Predicates and Quantifiers**	**(20 Hrs)**

Propositional Logic, Applications of Propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference

UNIT 2:	**Proof Methods**	**(10 Hrs)**

Introduction to Proofs, Proof Methods and Strategy.

UNIT 3:	**Mathematical Induction and Counting**	**(15 Hrs)**

Mathematical Induction, The Basics of Counting, The Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Identities.

Essential Reading:

K. H. Rosen, Discrete Mathematics and its Applications, 7th ed., McGraw – Hill, 2012.

Recommended Readings:

. R.P. Grimaldi and B.V. Ramana, Discrete and Combinatorial Mathematics, An applied introduction, 5th ed., Pearson Education, 2007.
. D. S. Chandrasekharaiah, Discrete Mathematical Structures, 4th ed., India: PRISM Book Pvt. Ltd., 2012.

. J. P. Tremblay and R. Manohar, Discrete Mathematical Structures with Application to Computer Science, Reprint, India: Tata McGraw Hill Education, 2008.

BCA102-1 OBJECT-ORIENTED PROGRAMMING USING C++

Total Hours for Semester: 75

Max Marks: 100 **Credits: 04**

Course Description:

This course provides a comprehensive introduction to Object-Oriented Programming (OOP) in C++. It covers fundamental concepts such as classes, objects, constructors, destructors, function and operator overloading, and various types of inheritance. The course also explores virtual functions, runtime polymorphism, and advanced file handling techniques. Practical lab exercises reinforce key programming concepts, ensuring hands-on experience with real-world applications.

Course Objectives:

- Understand and implement object-oriented programming concepts such as classes, objects, constructors, and destructors in C++.
- Develop proficiency in function and operator overloading, inheritance, and polymorphism to enhance code reusability and efficiency.
- Apply virtual functions, pointers, and dynamic memory allocation to create flexible programs.
- Gain hands-on experience in file handling, stream manipulation, and formatted I/O operations for effective data management in C++

Course Outcome

CO1: Understanding the principles of data abstraction, encapsulation, polymorphism and inheritance

CO2: Apply the principles of data abstraction, encapsulation

CO3: Apply the principles of polymorphism

CO4: Apply the principles of inheritance

CO5: Implementing the formatted and unformatted I/O

Unit I

Teaching Hours: 15

Classes and objects

Functions, arrays and strings in C++. Introduction - Defining a Class - Creating objects -Accessing class members-Defining member functions-Outside the class definition-Inside the class definition-Outside functions as inline-Nesting of member functions-Private member functions-Memory allocation for objects-Array of objects. Friend functions.

Lab Exercises:

1. Implementation of classes and objects.
2. Implementation of default arguments
3. Implementation of friend functions

Unit II

Teaching Hours: 15

Constructors and destructors

Basic Concepts of Constructors-Default Constructor-Parameterized Constructor-Multiple constructors in a Class-Constructor with default Arguments-Dynamic initialization of Objects-Copy Constructor-Dynamic Constructors-Destructors. Overloading Concepts Function Overloading: Functions with different sets of parameters, default and constant parameters.

Lab Exercises:

4 Implementation of default, parameterized and copy constructors
5 Implementation of static functions.
6 Implementation of Function overloading

Unit III

Teaching Hours: 15

Operator overloading

Rules for overloading Operators-Defining Operator Overloading-Overloading Unary Operators-Prefix and Postfix Operators Overloading-Overloading Binary Operators-Overloading using friend Functions,

Lab Exercises:-

7 Implementation of Unary operator overloading.

8 Implementation of Binary operator overloading.

Unit IV

Teaching Hours: 15

Inheritance

Introduction-Defining derived classes-Types of inheritances-Single-Making a private member inheritable-Multilevel Inheritance-Multiple Inheritance-Hierarchical inheritance-Hybrid inheritance-Virtual base classes-Abstract classes-Constructors in derived classes.

Virtual functions and run time polymorphism-Introduction-Compile time and Runtime polymorphism-Pointers to objects-this pointer-Pointer to derived classes-Virtual functions-Rules for virtual functions-Pure virtual functions.

Self-Learning

Hybrid inheritance

Lab Exercises:-

9 Implementation of multi-level inheritance.
10 Implementation of multiple inheritance

Unit V

Teaching Hours: 15

STREAMS

C++ stream classes-put and get functions-getline and write functions-Overloading << and >> operators-Formatted Console I/O operations-ios class functions-width, precision, fill, setf and unsetf-Formatting flags-Manipulators-User defined manipulators. Introduction to files in C++.

Self Learning

Put, get, width, precision

Essential Reading

[1] Schildt Herbert, *The Complete Reference C++*, Tata McGraw Hill, Reprint, 2018.

Recommended Reading

[1] Deitel&Deitel, *C++ How to program*, Pearson Education Asia, 6th Edition, 2016.

[2] Bjarne Stroustrup, *The C++ Programming Language,* Special Edition, Pearson Education, 2018.

[3] M. T. Somashekara, D. S. Guru, *Object-Oriented Programming with C++,* 2nd Edition, PHI, 2018.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	1	2	1	1	1
CO2	2	2	3	1	1	1	2	1

CO3	2	3	3	1	1	1	1	1
CO4	3	3	2	2	1	1	1	1

Web Resources:

https://www.w3schools.com

https://cplusplus.com/

Evaluation Pattern:

CIA - 50%
ESE 50%

BCA203-1 - PRINCIPLES OF SOFTWARE DEVELOPMENT and DBMS

Total Teaching Hours for Semester: 75

Max Marks: 100 **Credits: 4**

Course Description

This course provides a comprehensive introduction to fundamental software principles, including design methodologies and modelling techniques. It emphasizes core design concepts and software testing strategies, equipping students with a strong foundation in software development and quality assurance. Additionally, the course enhances students' understanding of strategic considerations and validation methodologies, fostering a deeper insight into software reliability and effectiveness.

Course Objectives

This course is an introduction to software development principles, process and design. And this course provides strong foundation of database concepts and develop skills for the design and implementation of a database application.

Course Outcomes

CO1: Design and Develop ethical software product
CO2: Apply terms, concepts, and tools of relational database management systems.
CO3: Construct database projects using DDL and DML queries.
CO4: Understand software design principles and apply design concepts like modularity, abstraction, and patterns.
CO5: Develop cases and requirement models to enhance software analysis and design.

Unit-1 **Teaching Hours: 15**

SOFTWARE ENGINEERING AND DBMS

Software Engineering: Nature of software- Defining software, Software Application Domains, Legacy Software - Software Engineering, The software process, Software Engineering practice. **Database Management System:** Data, Database, Database management system, Characteristics of the database approach, Role of Database administrators, Role of Database Designers, End Users, Advantages of Using a DBMS - Introduction to RDBMS- High-Level Conceptual Data Models for Database Design-Naming Conventions, and Design Issues.

Lab Exercises:

1. Introduction: logging on to SQL interface, exploring the system dummy tables.
2. Perform complex DDL, DML queries with where clause and violations.

Unit-2 **Teaching Hours:15**

DATA AND PROCESS MODEL

Process models : A generic process model-Process Assessment and improvement, Prescriptive Process Models – The waterfall Model, Incremental Model, Evolutionary Process Model, Concurrent Models. **Relationship Data Model :** Entity types, Entity Sets, Attributes and Keys. Relationships, Relationship types, Roles and Structural constraints. Weak Entity Types and Drawing E- R Diagrams – Relation, Integrity constraints - domain, entity and Referential integrity constraints: unique, not null, check, Primary Key, Foreign key.

Lab Exercise(s):

3. Demonstrate the use of constraints (UNIQUE, NOT NULL, CHECK, PRIMARY KEY, FOREIGN KEY) in SQL.
4. Write an SQL program to implement a Student Management System including the functionalities of ADD,VIEW,UPDATE and DELETE

Unit-3 **Teaching Hours:15**

SOFTWARE SYSTEM MODELING AND NORMALIZATION

Understanding Requirements : Requirements Engineering, Establishing the groundwork – Identifying Stakeholders, Recognizing multiple viewpoints, Working toward Collaboration, Usage Scenario Elicitation Work Products - Developing use cases, building the requirements model – Elements of the requirements Model. **Normalization, File Organization and Indexing:** Functional Dependencies - Normal Forms Based on Primary Keys-Second and Third Normal Forms-Boyce-Codd Normal Form - Multivalued Dependency and Fourth Normal Form-Join Dependencies and Fifth Normal Form-Inference Rules, Nulls and Dangling Tuples .

Lab Exercises:

5. Apply Integrity constraints using create and modify clauses.
6. Database Design using Normalization

Unit-4 **Teaching Hours:15**

DATA DESIGN AND PATTERN

Software Design and its Concepts: Design concepts – Abstraction, Architecture, Patterns, Separation of concerns, Modularity, information hiding, Functional Independence, refinement, Aspects, Refactoring, Object Oriented design concepts Design classes- Object-oriented design using Unified Modeling Language (UML)-Design Patterns -Test-driven development (TDD)-Code reviews and inspections. Enhanced Entity Relationship Model-Relational Database Design by EER-to-Relational Mapping and Inheritance. **SQL:**SQL data definition and data types, specifying constraints in SQL,

schema update statements, Basic queries, INSERT, DELETE and UPDATE statements in SQL. Views, Sub queries and correlated queries - Subclasses, Super classes.

Lab Exercises:

7. Create EER diagram with constraints.
8. Implement Sub queries.

Unit-5 **Teaching Hours:15**

SOFTWARE TESTING AND NoSQL

Software Testing Strategies : Strategic Issues - Development Testing – Test-driven development – **Basis Testing :** Release Testing – User testing – Availability and Reliability – Safety – Security Issues. – Validation Testing – System Testing – The Art of Debugging - **NoSQL Databases :** Use cases for No SQL Databases.

Self Learning : Software Testing tools – No SQL Basic Queries

Lab Exercise:

9. Design relation models (from the chosen domain)
10. Demonstrate a NoSQL program for basic CRUD operations

Text Books and Reference Books

[1] Software Engineering: A Practitioner's Approach" by Roger S. Pressman and Bruce R. Maxim: The most recent edition is the 9th edition, published in 2020 by McGraw Hill.

[2] "Fundamentals of Database Systems" by Ramez Elmasri and Shamkant B. Navathe: The latest edition is the 7th edition, published in 2016 by Pearson.

[3] "Professional NoSQL" by Shashank Tiwari: As of now, there is no newer edition beyond the original 2011 publication by John Wiley & Sons.

Essential Reading / Recommended Reading

[1] "Software Engineering" by Ian Sommerville: The most recent edition is the 10th edition, published in 2015 by Pearson.

[2]"Object-Oriented Modeling and Design" by James Rumbaugh: The latest edition is titled "Object-Oriented Modeling and Design with UML", co-authored by Michael Blaha and James Rumbaugh, published in 2004 by Pearson.

[3] "Database System Concepts" by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan: The most recent edition is the 7th edition, published in 2019 by McGraw Hill.

Web Resources:

1. Software Engineering, 9th Edition (futureuniversity.com)
2. https://www.nobledesktop.com/learn/software-engineering/free-resources-and-tutorials
3. https://www.techtarget.com/whatis/definition/software-engineering
4. https://www.mongodb.com/
5. https://www.ibm.com/cloud/learn/nosql-databases

CO – PO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	2	2	3	2	2	2	1	3
CO2	3	2	2	2	3	1	1	2
CO3	3	3	3	3	3	1	1	2
CO4	3	3	3	3	2	2	2	2
CO5	3	2	3	3	2	2	2	2

Evaluation Pattern:

CIA - 50%
ESE 50%

BCA161-1 - WEB TECHNOLOGIES

Total Teaching Hours for Semester: 60
Max Marks: 100 Credits: 03

Course Description

This course provides an in-depth introduction to the foundational technologies used in web development, covering HTML, CSS, JavaScript, PHP, and MySQL. Designed for beginners and intermediate learners, it equips students with the essential skills needed to create interactive, dynamic, and responsive websites.

Course Objectives

The objective of the course is to introduce the web page development technologies. In this course, students will be able to explore Hypertext Markup Language – its tags for content display and organization. CSS will be introduced for designing the web pages. PHP and MySQL will be introduced for Functionality and Database.

Course Outcomes

At the end of this course, students will be able to:

CO1 Apply HTML elements, attributes, and formatting techniques to design structured and visually appealing web pages.

CO2 Analyze and implement various CSS properties, including color schemes, layouts, and responsive web design, to enhance webpage aesthetics and usability.

CO3 Develop interactive web pages using JavaScript by incorporating variables, control structures, functions, and event handling.

CO4 Utilize PHP for server-side scripting by implementing variables, control structures, form handling, and database interactions.

CO5 Design and develop dynamic web applications by integrating MySQL databases for data storage, retrieval, and manipulation.

Unit-1 **Teaching Hours: 12**

Hypertext Markup Language (HTML)

Introduction to HTML Basic tags and Web Navigation, HTML Header, HTML Elements, HTML Attributes, Formatting – Comments - Images – Tables – Lists: Ordered and Unordered Lists - Test Links - Image Links – Frames – Blocks – Backgrounds – Colors: Text color, Background color,

Background color etc., – Fonts – Forms: Form elements – Embed multimedia – Marquees. Webpage layout.
Lab Exercises:
Program 01: Program to demonstrate HTML Header, Elements attributes and formatting
Program 02: Program to demonstrate Lists, Images, Background, Table and Forms

Unit-2 **Teaching Hours: 12**

Cascading Style Sheet (CSS)
CSS Introduction – CSS Versions – Syntax – Inclusion – Measurement units – CSS Color scheme – CSS Backgrounds for webpage and layout – CSS Fonts – CSS Text – CSS Image and related styles – CSS Links – CSS Tables – CSS Borders – CSS Margins – CSS Lists – CSS Padding – Cursors – Scrollbars; Advanced CSS: Positioning and Text effects, Media inclusion and types.
Self-study: CSS Responsive Web Designing

Lab Exercises:
Program 03: Program to demonstrate CSS color schemes, Text – Image – Links and table styles
Program 04: Program to demonstrate CSS lists, padding, and responsive designing.

Unit-3 **Teaching Hours: 12**

Java Script (JS Script)
Introduction to Java Script (Jscript) – Installation – Syntax – Variables – Operators – If else – Switch case – Loop controls: for loop, do while loop; Functions – Events – Cookies – Page redirect – Javascript Objects: Arrays, Date HTML DOM;
Lab Exercises:
Program 05: Program to demonstrate Jscript variables and operators
Program 06: Program to demonstrate loop controls – decision controls and functions

Unit-4 **Teaching Hours: 12**

Hypertext Preprocessor (PHP)
PHP Introduction – Installation version information – Environment setup – Variable types – Constants – Operator Types – Decision Making – Loop controls: for, do while; Arrays – Strings – Web concepts – GET and POST methods – Functions – Cookies -
Lab Exercises:
Program 07: Program to demonstrate PHP variable types – operators Decision and loop controls
Program 08: Program to demonstrate GET and POST methods

Unit-5 **Teaching Hours: 12**

Database Connection with MySQL
Introduction to MySQL, Installation version information – Connecting to MySQL database – Create a database – Delete a database – Insert data into MySQL Database: tables – Retrieve database from the database: tables – Updating data into the existing tables – Deleting data from the existing MySQL tables;
Lab Exercises:
Program 09: Program to demonstrate MySQL Database and Table creation and Insertion of data
Program 10: Program to demonstrate Updating data into the existing tables

Text Books and Reference Books

1. Thomas A. Powell, “HTML & CSS: The Complete Reference, Fifth Edition”, McGraw Hill LLC, Thomas A. Powell, “HTML The Complete Reference”, Osborne/McGraw-Hill.
2. David Flanagan, “JavaScript - The Definitive Guide”, O'Reilly Media.
3. Rasmus Lerdorf, Kevin Tatroe, Bob Kaehms, Ric McGredy, Nathan Torkington, Paula M. Ferguson, “Programming PHP”, O'Reilly Media
4. Saied M.M. Tahaghoghi, Hugh E. Williams, “Learning MySQL”, O'Reilly Media.

Essential Reading / Recommended Reading

1. Jon Duckett, “HTML and CSS - Design and Build Websites”, Wiley, ISBN: 9781118008188, 1118008189.
2. Bryan Pfaffenberger, “HTML, XHTML, and CSS Bible”, Wiley Pub, ISBN: 9788126505401.
3. Eric A. Meyer, “CSS Pocket Reference”, O'Reilly Media, ISBN: 9781449313050.
4. Axel Rauschmayer, “Speaking JavaScript - An In-Depth Guide for Programmers”, O'Reilly Media, ISBN:9781449364991.
5. Addy Osmani, “Learning JavaScript Design Patterns”, O'Reilly Media, ISBN: 9781449334871.
6. Josh Lockhart, “Modern PHP - New Features and Good Practices”, O'Reilly Media, ISBN: 9781491904992.
7. Matt Doyle, “Beginning PHP 5.3”, Wiley, ISBN: 9781118057346.
8. David Sklar, Adam Trachtenberg, “PHP Cookbook”, 3rd Edition, O'Reilly Media, Inc., ISBN: 9781449363758.

Web Resources:

1. https://developer.mozilla.org/en-US/
2. https://www.theodinproject.com/

3. https://www.freecodecamp.org/

CO – PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	2	1	3	1	1	1
CO2	3	3	3	2	3	1	1	1
CO3	3	3	3	3	3	1	1	1
CO4	3	3	3	3	3	2	1	1
CO5	3	3	3	3	3	2	1	1

Evaluation Pattern:

CIA - 50%
ESE 50%

SEMESTER 2

Course Name :	**DISCRETE MATHEMATICS**	**Course Code:**	**BCA104-2**
Course Duration :	**45**	**Credits:**	**3**
Course Description: This course aims at introducing the students into the world of Discrete Mathematics. It includes topics like Set Theory, Functions, Matrices, Relations, and Number theory. They gain a historical perspective of the development of modern discrete mathematics and application of the same in the field of Computer Science.			
Course Objectives: This course will help the learner to			

COBJ1. be proficient in the topics set theory, functions, matrices, relations and number theory.

COBJ2. enhance the problems solving skills in set theory, functions, matrices, relations and number theory.

Course Outcomes: On successful completion of the course, the students should be able to

CO1. demonstrate a working knowledge of set notation and elementary set theory.

CO2. apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3. understand different types of matrices and operations among matrices.

CO4. determine when a relation is reflexive, symmetric, antisymmetric, or transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5. effectively express the elementary concepts and results of Number theory

Pedagogy: Lectures, discussions, problem solving and seminars

Syllabus:

UNIT 1:	**Sets, Functions and Matrices**	**(15 Hrs)**

Sets, Set Operations, Functions, types of functions, Matrices and Matrix operations.

UNIT 2:	**Relations**	**(10 Hrs)**

Relations and Their Properties, Equivalence Relations, Partial Orderings

UNIT 3:	**Number Theory**	**(20 Hrs)**

Divisibility and Modular Arithmetic, Integer representations, primes and greatest common divisors, solving congruences and its applications.

Essential Reading:

K. H. Rosen, Discrete Mathematics and its Applications, 7th ed., McGraw – Hill, 2012.

Recommended Readings:

. R.P. Grimaldi and B.V. Ramana, Discrete and Combinatorial Mathematics, An applied introduction, 5th ed., Pearson Education, 2007.
. D. S. Chandrasekharaiah, Discrete Mathematical Structures, 4th ed., India: PRISM Book Pvt. Ltd., 2012
. J. P. Tremblay and R. Manohar, Discrete Mathematical Structures with Application to Computer Science, Reprint, India: Tata McGraw Hill Education, 2008.

BCA105-2 OPERATING SYSTEM

Duration: **60 Hours** **Max Marks : 100**

Credits: 4

Course Description

This course covers essential topics such as process management, memory management, file systems, and device management, along with advanced concepts like CPU scheduling algorithms, synchronization mechanisms, and deadlock handling. Students gain insights into the architecture and functionality of modern operating systems and explore practical implementation aspects through hands-on labs and real-world examples. The course aims to equip students with the skills needed to design, analyze, and optimize operating system functionalities, preparing them for careers in software development, system administration, and research in computer science.

Course Objectives

To provide comprehensive knowledge of operating system concepts, including system structure, process management, deadlocks, memory management, and file systems, while offering practical exposure to implementing these concepts using C/C++.

Course Outcomes

CO1: Understand the fundamental concepts, components, and services of an operating system.
CO2: Apply process scheduling algorithms and memory management techniques to solve real-world computing problems.
CO3: Analyze file system structures, disk scheduling algorithms, and storage management strategies for efficient data handling.
CO4: Evaluate security mechanisms in modern operating systems and propose solutions to mitigate security threats.

CO5: Investigate operating system security and protection mechanisms

Unit 1: Introduction to Operating Systems 12 Hours

Definition and Importance of Operating Systems-Objectives and Functions of Operating Systems-Types of Operating Systems: Batch, Time-Sharing, Distributed, Real-Time, and Embedded-Components of an Operating System: Kernel, Shell, and System Programs -Overview of Operating System Services - System Calls: Definition, types, and example.

Unit 2: Process Management 12 Hours

Concept of Processes: Process states, Process Control Block (PCB) -Threads: Definition, benefits, and multithreading models -Process Scheduling: Long-term, short-term, and medium-term scheduling-CPU Scheduling Algorithms: First Come First Serve (FCFS), Shortest Job Next (SJN), Priority Scheduling, Round Robin (RR) -Inter-process Communication (IPC): Message passing and shared

memory -Process Synchronization: Critical Section Problem, Semaphores, Monitors.Deadlock prevention and detection.

Unit 3: Memory Management 12 Hours

Memory Hierarchy: Cache, RAM, Virtual Memory -Contiguous Memory Allocation: Fixed and variable partitioning, fragmentation -Paging and Segmentation: Concepts, paging tables, and segment tables - Virtual Memory: Demand paging, page replacement algorithms (FIFO, LRU)

Unit 4: File Systems and Storage Management 12 Hours

File Systems: File attributes, operations, and types - Directory Structures: Single-level, two-level, tree-structured, acyclic-graph - File Allocation Methods: Contiguous, linked, indexed - Free Space Management: Bitmaps, linked lists, grouping, and counting Disk Scheduling: FCFS, SSTF, SCAN, C-SCAN algorithms

Unit 5: Security and Protection 12 Hours

Basics of Security and Protection in Operating Systems -Access Control: Access Control Lists (ACLs), Role-Based Access Control (RBAC) -Authentication Methods: Passwords, tokens, biometrics -Common Threats: Malware, viruses, worms, and spyware - Security Measures in Modern Operating Systems (Linux and Windows) Case Study: Securing an operating system.

Case Study(Self-Learning)- Introduction to Linux and Windows Operating Systems.

Textbooks

1. **Abraham Silberschatz, Peter B. Galvin, Greg Gagne**, *Operating System Concepts*, 10th Edition, Wiley, 2018.
2. **Andrew S. Tanenbaum, Herbert Bos**, *Modern Operating Systems*, 4th Edition, Pearson, 2023.
3. **William Stallings**, *Operating Systems: Internals and Design Principles*, 9th Edition, Pearson, 2022.

Reference Books

1. **Dhananjay M. Dhamdhere**, *Operating Systems: A Concept-Based Approach*, 3rd Edition, McGraw-Hill, 2022.
2. **Thomas Anderson, Michael Dahlin**, *Operating Systems: Principles and Practice*, 2nd Edition, Recursive Books, 2021.
3. **Charles Crowley**, *Operating Systems: A Design-Oriented Approach*, McGraw-Hill, Latest Edition.
4. **Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau**, *Operating Systems: Three Easy Pieces*, Self-Published, 2023 (Available Online).

CO-PO mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	2	1	1	1	1
CO2	3	3	2	2	2	1	1	1
CO3	3	3	3	3	3	2	2	2
CO4	3	2	3	3	3	3	3	2
CO5	1	2	3	1	2	3	1	2

Evaluation Pattern:

CIA - 50%
ESE - 50%

SEMESTER 2

BCA206-2 – DATA STRUCTURES USING C++

Total Teaching Hours for Semester: 75

Max Marks: 100 **Credits: 4**

Course Description

This course provides an in-depth understanding of fundamental data structures and algorithms using C++. The course covers essential concepts of vectors, arrays, stacks, queues, linked lists, trees, and sorting/searching techniques. Practical implementations using pointers, classes, and abstract data types are emphasized through structured lab exercises.

Course Objectives

This course provides a basic concept of Abstract Data Types, Linear and Non-Linear Data structures, and the behaviour of data structures such as stacks, queues, trees, hash tables, search trees, Graphs and their representations. This Course also analyse various searching and sorting algorithms and also implements ADTs such as lists, stacks, queues, trees, search trees in C++ to solve problems.

Course Outcomes:

CO1: Discuss vector and array data structures, as well as efficient usage of pointers and class objects.

CO2: Implement stack and queue data structures, along with applications such as expression conversion and evaluation.

CO3: Analyse linked list operations for singly, circular, and doubly linked lists, enhancing data organization techniques.

CO4: Evaluate algorithms for searching and sorting using various methods such as linear search, binary search, and quick sort.

CO5: Create binary tree operations, including insertion, deletion, and various traversal techniques.

Unit I **Teaching Hours: 15**

DATA STRUCTURE INTRODUCTION: Introduction - Linear Vs Non-Linear Data Structure - Memory, Addresses and Pointers - **Arrays**: Array Initialization - Pointers and Arrays . **Vector:** Vector Basics - Pointers to Class Objects. **Structure:** Declaration - Structure using Arrays -Structure using Pointers. Problem Solving: Sparse Matrix- Palindrome using Arrays and Pointers.

Lab Exercises:

1. Implement Sparse Matrix using Arrays
2. Implement Palindrome using Pointers or Vectors

Unit-II **Teaching Hours: 15**

STACKS AND QUEUES: Stack ADT: definition, Stack Implementation, Infix to postfix conversion, Postfix expression evaluation, recursion implementation, **Queue ADT:** definition, Queue Implementation.

Lab Exercises:

1. Write a C++ program to implement a Stack using ADT
2. Write a C++ program to implement a Queue using ADT

Unit-III **Teaching Hours: 15**

LINKED LIST: Linear list ADT-Linked representation, Singly Linked Lists Operations - Insertion, Deletion, Circular linked lists, Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion, Deletion.

Lab Exercises:

1. Write a C++ programs to implement singly list ADT to perform following operations
    a. Insert an element into a list.
    b. Delete an element from list
    c. Search for a key element in list
    d. count number of nodes in list
2. Write a C++ programs to implement Doubly list ADT to perform following operations
    a. Insert an element into a list.
    a. Delete an element from list
    b. Search for a key element in list
    c. count number of nodes in list

Unit-IV **Teaching Hours:15**

SEARCHING AND SORTING: Searching : Linear Search, Binary Search. **Sorting:** Insertion Sort, Selection Sort, Bubble Sort, Quick sort, Heap Sort, Comparison of Sorting methods.

Lab Exercises:

1. Implement Linear Search and Binary search
2. Implement Quick sort and Selection sort

Unit-V **Teaching Hours:15**

TREES: Trees: Introduction, Terminology, Representation of Trees, Binary Trees, The Abstract Data Type, Properties of Binary Trees, Binary Tree Traversal. Introduction, Inorder Traversal, Preorder Traversal, Postorder Traversal. **Binary Search Trees**: Definition, Implementation, Operations- Searching, Insertion and Deletion.

Lab Exercises:

1. Implement the below operations to Binary Tree.
    i)Insertion
    ii)Traverse a given binary tree in the following ways:
        Inorder Traversal (Left, Root, Right)

Preorder Traversal (Root, Left, Right)
Postorder Traversal (Left, Right, Root)

iii)Deletion

iv)Display

Text Books and Reference Books

1. Programming: Principles and Practice Using C++, Bjarne Stroustrup , Addison- Wesley Publication, 2nd Edtion, 2014 (Unit I)
2. Data structures, Algorithms and Applications in C++, S.Sahni, University Press (India) Pvt.Ltd, 2nd edition, Universities Press, Pvt. Ltd. 2005 (Unit II,III,IV, and V)

Essential Reading / Recommended Reading

1. Data structures and algorithms in C++, Adam Drozdek, Thomson, 3rd Edition, 2012
2. Data Structure and Algorithms Using C++ - A Practical Implementation, By Sachi Nandan Mohanty, Pabitra Kumar Tripathy, WILEY, · 2021
3. Data structures and Algorithms in C++, Michael T.Goodrich, R.Tamassia and .Mount, John Wiley and Sons, Wiley student edition, 2011

Web Resources

1. **dl.icdst.org/pdfs/files3/fef0590f02fa06bb42cba558fbc9e51c.pdf**
2. **Data Structure and Algorithms Using C++ - Google Books**
3. **Data Structures and Algorithms in C++ - Google Books**

CO – PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	1	1				1	
CO2	2	3	2		2		1	1
CO3	3	1	3				1	1
CO4	2	3	3	1			1	1
CO5	3	2	3		1		1	1

Evaluation Pattern

CIA - 50%

ESE - 50%

BCA262-2: WEB STACK DEVELOPMENT

Total Teaching Hours for Semester: 60
Max Marks: 100 **Credits: 3**

Course Description

The course covers the end-to-end development process, from the front-end to the back-end, and includes an introduction to web frameworks, databases, APIs, and deployment strategies.

Course Objective

This course gives an essence of web developer technologies and to develop web applications using advanced technologies and cultivate good web programming style and discipline by solving the real-world scenarios.

Course Outcomes

CO1: Apply AJAX concepts to create dynamic web pages with efficient data loading, asynchronous behavior, and error handling.
CO2: Develop responsive and interactive front-end applications using React JS, implementing components, hooks, routing, and state management techniques.
CO3: Utilize JavaScript and AngularJS for client-side scripting, handling events, manipulating the DOM, validating forms, and creating single-page applications.
CO4: Implement server-side scripting using Node.js, managing file systems, debugging applications, and handling streams and events efficiently.
CO5: Integrate Node.js with MySQL to perform database operations, handle data securely, and implement file uploads and retrieval functionalities.

Unit-1 **Teaching Hours: 12**

AJAX

AJAX and page reloading - Basic structure of AJAX request – Http Request object-Sending GET and POST requests-Handling server responses and status codes-Asynchronous behaviour and callbacks-Error handling in AJAX-jQuery AJAX methods-Common AJAX errors-Implementing user feedback for AJAX errors

self-learning: Basic structure and styling of web pages- Understanding of functions, event handling, and DOM manipulation - Data format commonly used in AJAX responses-Understanding GET, POST, status codes, and request/response cycle.

Lab Exercises:

1. Create dynamic page with data loading via AJAX
2. Create a form with error handling and AJAX status feedback

Unit-2 **Teaching Hours: 12**

React JS

Package Manager (NPM) - ES6- Introduction to React.js - Create React App & React file structure - JSX and Components -passing and destructuring props - React Hooks - Axios - Images and Forms - Conditional Rendering - Routes - Redux

Lab Exercises:

3. Create a web application using React JS with Forms.
4. Develop SPA (Single Page Application) with React JS.

Unit-3 **Teaching Hours: 12**

CLIENT-SIDE SCRIPTING

JavaScript Implementation - Use JavaScript to interact with some of the new HTML5 APIs -Create and modify JavaScript objects- JS Forms - Events and Event handling-JS Navigator-JS Cookies Introduction to JSON-JSON vs XML-JSON Objects-Importance of Angular JS in web Angular Expression and Directives-Single Page Application.

Lab Exercises:

1. JavaScript program to demonstrate Form Validation and Event Handling.
2. Create a web application using AngularJS with Forms.

Unit-4 **Teaching Hours: 12**

SERVER-SIDE SCRIPTING

Introduction to Node.js-REPL Terminal-Package Manager (NPM)-Node.js Modules and filesystemNode.js Events-Debugging Node JS Application-File System and streams-Testing Node JS with jasmine.

Lab Exercises:

1. Demonstrate Node.js file system module.
2. Demonstrate File System and Streams.

Unit-5 **Teaching Hours: 12**

Node JS and MYSQL

Introduction to Node.js - Express JS - Node mailer - Node JS with MYSQL - Introduction to MySQL - Performing basic database operation (DML) (Insert, Delete, Update, Select)-Prepared Statement-Uploading Image or File to MySQL- Retrieve Image or File from MySQL.

Lab Exercises:

1. Demonstrate Node.js file system module.
2. Implement CRUD operation with MySQL using Node.JS

Text Books and Reference Books

1. Paul Deitel , Harvey Deitel & Abbey Deitel, “Internet and World Wide Web: How to Program”, Pearson Education, 5th Edition, 2018.

2. Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 1st Edition, 2020
3. Stephen Burge, "MSQL Explained: Your Step-by-Step Guide to Database Design", CreateSpace Independent Publishing Platform, 2017.

Essential Reading / Recommended Reading

1. Chris Northwood, The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer, Apress Publications, 1st Edition, 2018.
2. Laura Lemay, Rafe Colburn & Jennifer Kyrnin, Mastering HTML, CSS & Javascript Web Publishing, BPB Publications, 1st Edition, 2016.

Web Resources:

1. https://html.com/html5/
2. https://react.dev/
3. https://angularjs.org/
4. https://nodejs.org/en

CO – PO Mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8
CO 1	3	3	3	2	2	2	-	2
CO 2	2	3	3	2	1	2	-	2
CO 3	1	2	3	3	2	3	-	1

Evaluation Pattern

CIA - 50%

ESE - 50%

SEMESTER 3

STA306-3 STATISTICS FOR COMPUTER APPLICATIONS

Course Name:	**STATISTICS FOR COMPUTER APPLICATIONS**	**Code:**	**STA306-3**
Total Hours:	**45**	**L+P+T:**	**3+0+0**
Max Marks:	**100**	**Credits:**	**3**
Course Type:	**Major Core**	**Course Category:**	**Theory**

-Course Description

The course emphasizes applying statistical methods to solve scientific, engineering, and real-world problems for effective decision-making.

Course Objectives

This course aims to teach students the fundamental techniques of statistical methods. After completing the course, students will be able to apply these methods to solve problems in financial, scientific, and engineering fields.

Course Outcomes

CO1: Explain the concepts of data collection, classification, tabulation, and graphical representation of data.

CO2: Determine measures of central tendency for individual, discrete, and continuous series.

CO3: Analyze measures of dispersion, standard deviation, and coefficient of variation.

CO4: Apply correlation techniques such as Karl Pearson's and Spearman's methods to study relationships between variables.

CO5: Apply fundamental probability concepts such as conditional probability and Bayes' theorem to solve problems.

Unit-1 **Teaching Hours: 5**

Introduction

Meaning and Definition of Statistics – Collection of data — Primary and Secondary - Classification and Tabulation – Diagrammatic and Graphical presentation.

Unit-2 **Teaching Hours: 12**

Measures of Central Tendency

Simple Arithmetic Mean - Arithmetic Mean in case of Individual Series - Discrete series and continuous series - Weighted Arithmetic Mean - Median in case of Individual - Discrete series and continuous series - Mode in case of Individual Series - Discrete series and continuous series.

Unit-3 **Teaching Hours: 12**

Measures of Dispersion

Computation of Range - Mean Deviation – Variance - Standard Deviation for individual Series - Discrete Series and Continuous Series - Coefficient of Standard Deviation and coefficient of variation - Combined Standard Deviation - Correcting incorrect Standard Deviation.

Unit-4 **Teaching Hours: 8**

Correlation

Correlation – Meaning and Definition – Scatter diagram, Karl Pearson's co-efficient of Correlation, Spearman's Rank Correlation,

Unit-5 **Teaching Hours: 8**

Basics of Probability

Introduction to Probability - Sample Space and Events - Definition of Probability - Axiomatic approach to probability - Addition Law of Probability - Conditional Probability - Bayes' Theorem

Text & Reference Books

[1] Gupta S.C and Kapoor V.K, Fundamentals of Mathematical Statistics, 12th edition, Sultan Chand & Sons, New Delhi, 2020.

[2] Mukhopadhyay P, Mathematical Statistics, Books and Allied (P) Ltd, Kolkata, 2018.

Essential Reading

[1] Walpole R.E, Myers R.H, and Myers S.L, Probability and Statistics for Engineers and Scientists, Pearson, New Delhi, 2017.

[2] Montgomery D.C and Runger G.C, Applied Statistics and Probability for Engineers, 7th Edition, Wiley India, New Delhi, 2018.

[3] Agarwal B.L, Basic Statistics, 6th Edition, New Age International (P) Limited Publishers, 2018

Additional Information (Web Resources):

1. Preface - Introductory Statistics | OpenStax
2. Introduction to Statistics | InfoBooks.org

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	N,G	3	2	1	–	–	1	1	–
CO2	N,G	3	3	2	1	–	1	–	–

CO3	N,G	3	3	2	1	–	1	–	–
CO4	N,G	2	3	2	1	–	1	1	–
CO5	N,G	2	3	2	2	1	1	–	–

Evaluation Pattern: CIA - 50% ESE - 50%

INDUSTRY INTEGRATED COURSE

BCA207A-3– ROBOTIC PROCESS AUTOMATION

Course Name:	**ROBOTIC PROCESS AUTOMATION**	**Code:**	**BCA207A-3**
Total Hours:	**45**	**L+P+T:**	**1+2+0**
Max Marks:	**100**	**Credits:**	**2**
Course Type:	**Major Core**	**Course Category:**	**Theory and Practical**

Course Description

This industry-integrated course, designed in collaboration with UiPath Academic Alliance, equips students with hands-on experience in RPA development. Learners will gain expertise in automation tools, control flow, error handling, data manipulation, UI automation, and project management. The course also prepares students for UiPath RPA certification and includes hands-on labs and industry expert sessions.

Course Objectives

The Automation Design and Development curriculum is designed to build automation development competency from the ground up while also diving into more broader areas to gain a holistic understanding of how automation intertwines with various technologies.

Course Outcomes

CO1: Explain and utilize the fundamentals of Robotic Process Automation

CO2: Develop automation workflows using UiPath Studio

CO3: Investigate and implement UI automation techniques for efficient automation solutions

CO4: Develop the ability to independently design and create robots for business processes

CO5: Deploy and manage automation projects using UiPath Orchestrator

Unit-1 **Teaching Hours: 9**

INTRODUCTION TO ROBOTIC PROCESS AUTOMATION (RPA) CONCEPTS, TOOLS

Robotic Process Automation (RPA) – Benefits - UiPath Business Automation Platform - Core RPA components: Studio, Orchestrator and Robot with Assistant – Types of UiPath robots: Attended and Unattended – Key Components of the UiPath Studio user interface - modern vs classic design - Create

and configure variables in an automation project – Working with variables, arguments, and workflow files – Global constants and global variables - Arrays

1. Creating a Simple Automation Workflow
2. Working with Variables, Arguments, and Arrays

Unit-2 **Teaching Hours: 9**

CONTROL FLOW, COMMON RPA IMPLEMENTATIONS, EXCEPTION HANDLING AND DEBUGGING

Sequences and Flowcharts – Control Flow Statements for Decisions and Iterations – Applying Excel and Workbook Activities for Automation – Types of Exceptions in Automation – Best Practices for Error and Exception Handling – Selecting, Creating, Deleting, Moving, and Renaming Files and Folders – Accessing and Working with Gmail using UiPath Studio – Implementing Microsoft Office 365 Activities for Automation – Extracting Large Text Segments and Single Pieces of Information from PDF Files

3. Automating Excel Data Processing
4. Email Automation using UiPath

Unit-3 **Teaching Hours: 9**

DATA MANIPULATION AND VARIABLES IN AUTOMATION

Data manipulation – Use the most common VB.NET methods to manipulate string variables – Perform complex string manipulation using RegEx builder – String Manipulation using string activities – Lists – Features of Invoke and String Conversion methods – Differences between Arrays and Lists – Dictionary variables – Data table variable – Worksheet Vs a data table.

5. Automation using String Manipulation and Regular Expressions
6. Working with Data Tables

Unit-4 **Teaching Hours: 9**

UI AUTOMATION, DESCRIPTORS AND SELECTORS

Synchronizing Activities in Automation Workflows – Check App State activity and Verify Execution feature – Pick Branch Activity – Different Targeting Methods – Validate target elements –Dynamic Text Target option, wildcards, variables – Selectors: Structure and type - Property Explorer tool

7. Automating Web Data Extraction

8. Dynamic Selectors and App State Verification

Unit-5 **Teaching Hours: 9**

ORCHESTRATOR AND PROJECT MANAGEMENT IN AUTOMATION

Orchestrator capabilities and entities – Tenant Context vs Folder Context – Use Orchestrator resources directly in Studio – Version Control Systems: Basic GIT features, Benefits and Challenges – Project Layout for an Automation Process – Break down a complex automation project into functional workflows – Benefits of utilizing best practices for project organization

9. Deploying and Managing Bots in Orchestrator

10. Using GIT for Version Control in UiPath

Text & Reference Books

[1] Ramesh Nougain, *Robotic Process Automation for Developers*, Packt Publishing, 978-1789139488, 2019.

[2] Kris P. K. Kothari, *Mastering UiPath: A Comprehensive Guide to Building RPA Applications with UiPath*, Wiley, 978-1119686979, 2020

Essential Reading

[1] Srikanth Merianda,"Robotic Process Automation Tools, Process Automation and their benefits: Understanding RPA and Intelligent Automation", Consulting Opportunity Holdings LLC, 1st Edition 2018.

Additional Information (Web Resources):

1. https://docs.uipath.com/studio/standalone/2023.10/user-guide/install-studio
2. https://docs.uipath.com/

3. https://community.uipath.com/
4. https://forum.uipath.com/
5. https://academy.uipath.com/learning-plans/automation-developer-associate-training-(v2023.10)
6. https://www.uipath.com/learning/certification

CO – PO Mapping

	Development Needs (LRNG)	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**P08**
CO1	NG	3	1	2	2	1	1	2	1
CO2	NG	2	2	3	3	2	2	1	2
CO3	NG	2	3	3	2	3	2	1	3
CO4	NG	3	2	3	2	2	3	2	2
CO5	NG	2	2	3	2	1	3	2	3

Evaluation Pattern: CIA - 50% ESE - 50%

GENERATIVE AI TOOLS AND CREATIVE COMPUTING

Course Name	**Generative AI Tools and Creative Computing**	**Course Code**	**BCA207C-3**
Total Hours	**45**	**L+P+T**	**1+2+0**
Max Marks	**100**	**Credits**	**2**
Course Type	**Major Course**	**Course Category**	**Theory + Practical**

Course Description

This industry-integrated course introduces students to the concepts and applications of Generative Artificial Intelligence with a strong emphasis on creative computing and productivity tools. The course enables learners to utilise AI-powered tools for text, image, and content generation, while also providing hands-on exposure to executing Generative AI models on high-performance computing infrastructure. Students will gain practical experience with AI infrastructure for running AI models, understanding computational workflows, and applying AI across interdisciplinary contexts.

Course Objectives

The course is designed to:

- Introduce foundational concepts of Generative AI without requiring prior Machine Learning/Deep Learning knowledge
- Enable students to effectively use AI tools for creative and academic purposes
- Provide hands-on exposure to AI model execution on server-based infrastructure
- Develop prompt engineering and AI-assisted problem-solving skills
- Promote ethical and responsible use of AI technologies

Course Outcomes

CO1: Explain the fundamentals and applications of Generative AI and creative computing

CO2: Apply prompt engineering techniques to generate effective and context-aware outputs

CO3: Utilize Generative AI tools for content creation, design, and productivity tasks

CO4: Execute and integrate AI workflows using AI infrastructure

CO5: Analyze ethical issues, bias, and responsible usage in AI-driven solutions

Unit – 1 **Teaching Hours: 9**

INTRODUCTION TO GENERATIVE AI AND PROMPT ENGINEERING

Introduction to Artificial Intelligence – Evolution of AI – Overview of Generative AI – Types of Generative AI (Text, Image, Code) – Applications of Generative AI across domains (education, business, media, healthcare).

Introduction to prompt engineering – Types of prompts – Context-based prompting – Role-based prompting – Iterative prompting and refinement – Prompt optimization techniques

Lab Exercises:

1. Designing Structured Prompts for Different Use Cases
2. Comparing Outputs from Different Generative AI Tools
3. Exploring AI Tools for Text and Content Generation using Online Tools and other AI Infrastructure

Unit – 2 **Teaching Hours: 9**

AI TOOLS AND CREATIVE COMPUTING

Overview of popular AI tools and platforms – AI-assisted productivity and creativity – Limitations of Generative AI systems.

AI-assisted content creation (blogs, reports, presentations) – AI for image generation and design – AI-assisted coding tools – Creative applications of Generative AI

Lab Exercises:

4. Basic Linux Commands and SSH Access for Server Interaction
5. Accessing Server and Running Basic Generative AI Models using Jupyter Environment

Unit – 3 **Teaching Hours: 9**

GENERATIVE AI USING AI INFRASTRUCTURE

Need for high-performance computing in AI – Overview of Intel Xeon architecture (conceptual) – CPU vs GPU in AI workloads – Introduction to server-based computing – Accessing Lab infrastructure (SSH, Jupyter Notebook) – Running pre-trained Generative AI models – Text generation using local models – Handling datasets and files in server environment – Basic AI execution workflows

Lab Exercises:

6. Processing Input Data and Executing Prompt-Based Tasks using Generative AI Models on AI Infrastructure
7. Generating Text Outputs using Local Generative AI Models on AI Infrastructure

Unit – 4 **Teaching Hours: 9**

AI WORKFLOWS, MULTI-MODAL SYSTEMS AND APPLICATION DEVELOPMENT

End-to-end AI workflow design – Integrating multiple AI tools – Processing large inputs using server-based systems – Batch processing using AI models – Multi-modal AI systems (Text to Image, Image to Text, Text to Content) – Domain-specific AI applications (education, business, media) – AI-driven automation workflows – Introduction to reusable AI scripts and basic deployment concepts

Lab Exercises:

8. AI-Powered Document Processing (Summarization and Key Points Extraction) using AI Infrastructure
9. End-to-End Generative AI Application Development integrating multiple tools and METIS execution

Unit – 5 **Teaching Hours: 9**

RESPONSIBLE AI AND MINI PROJECT

AI bias and fairness – Data privacy and security – Copyright and intellectual property in AI-generated content – Risks and misuse of Generative AI – Academic integrity and ethical AI usage – Case studies on responsible AI – Guidelines for ethical AI development and deployment

Lab Exercises:

10. Mini Project

Design and develop an **end-to-end Generative AI solution** using AI tools and AI infrastructure – Problem identification and solution design – Implementation using prompt engineering and AI tools – Execution on Intel server – Output generation and documentation – Workflow representation

Ethical Analysis:
Identification of bias in AI outputs – Evaluation of accuracy and limitations – Data privacy concerns – Misuse scenarios – Suggested mitigation strategies

Text & Reference Books

[1] Bernard Marr, *Generative AI in Practice*, Wiley, 2023
[2] David Foster, *Generative Deep Learning*, O'Reilly (Conceptual Reference)

Essential Reading

[1] OpenAI Documentation and AI Usage Guidelines
[2] Intel oneAPI AI Analytics Toolkit Documentation

Additional Web Resources

1. https://www.intel.com/oneapi
2. https://huggingface.co/models
3. https://platform.openai.com/docs
4. https://www.tensorflow.org/resources
5. https://pytorch.org/tutorials

CO – PO Mapping

COs \ POs	(LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	NG	3	2	1	2	1	2	1	1
CO2	NG	2	3	2	2	2	2	2	1
CO3	NG	2	2	3	3	3	2	2	1
CO4	NG	3	2	3	3	2	2	2	1
CO5	NG	2	3	2	2	2	3	2	3

Evaluation Pattern: CIA - 50% ESE - 50%

BCA208-3 – PROGRAMMING WITH JAVA

Course Name:	**Programming With Java**	**Code:**	**BCA208-3**
Total Hours:	**75**	**L+P+T:**	**3+2+0**
Max Marks:	**100**	**Credits:**	**4**
Course Type:	**Major Core**	**Course Category:**	**Theory and Practical (LCA)**

Course Description

This course provides a comprehensive understanding of Java programming, beginning with object-oriented principles, core syntax, and fundamental programming constructs. It further covers advanced concepts such as collections, generics, lambda expressions, GUI development, database connectivity, and web technologies including Servlets and JSP. Emphasis is placed on practical implementation, scalability, and adherence to industry standards to prepare students for real-world software development.

Course Objectives

The objective of this course is to impart a strong understanding of object-oriented principles and advanced Java technologies. The practical sessions integrated with the course aim to make students industry-ready by enabling them to design, develop, and test Java applications. This course helps learners gain sound knowledge of advanced Java concepts through hands-on programming experience.

Course Outcomes

CO1. Understand and apply fundamental Java programming concepts to develop structured and modular programs.

CO2. Apply object-oriented programming concepts to design modular, reusable, and maintainable Java applications.

CO3. Apply Java collection framework, generics, file handling, lambda expressions, and Stream API to process data efficiently and develop maintainable and performance-oriented applications.

CO4. Develop desktop-based GUI applications using Swing, following MVC principles and integrate database connectivity using JDBC

CO5. Analyze and develop Java-based web applications using Servlets and JSP by understanding request–response handling and web application architecture.

Unit-1 **Teaching Hours: 15**

ARRAYS and CLASSES

Arrays: JAVA Features –Bytecode – Declaring Arrays - Accessing Array Elements – For Each loop – Array Copying–Command-Line Parameters – Multi-Dimensional Arrays – Ragged Arrays

Classes and Objects: General Form of a Class- Declaring Objects – Introducing Methods – Parameterised Methods-Method Overloading – Using Objects as Parameters – Constructors – Parameterized Constructors - Constructor Overloading – this keyword – Garbage collection – Access Control – static keyword - Relationship Between Classes – Using Predefined Classes- Mutator and Accessor Methods

Self Learning:

Introduction to Java Coding Standards

Lab Exercises:

1. Implement the concept of class, data members, member functions and access specifiers in the chosen domain.
2. Implement array of objects with constructor and function overloading

Unit-2 **Teaching Hours: 15**

INHERITANCE, PACKAGES, INTERFACES, EXCEPTION HANDLING and MULTITHREADING

Inheritance, Packages and Interfaces: Super Classes and Subclasses - Defining Subclasses – Overriding Methods- Dynamic Method Dispatch - Subclass constructors – Inheritance hierarchies- Polymorphism- Understanding method calls- final Classes and Methods – Casting – Abstract Classes–Interfaces– Packages.

Exception Handling and Multithreading: Exception handling– Exception types –uncaught exception – Using try, catch and finally – Multiple catch clauses- Nested try statements – throw – throws – Built in exception – User defined exception – Thread Model – Life cycle of a Thread - Java Thread Priorities – Runnable interface and Thread Class –Thread Synchronization– Inter Thread Communication.

Lab Exercises:

3. Implement the concept of inheritance, package and interface.
4. Implement the concept of exception handling and multithreading concepts.

Self Learning

Type casting and synchronization challenges in Java

Unit-3 **Teaching Hours: 15**

COLLECTION, GENERICS, FILE HANDLING, LAMBDA EXPRESSION and STREAMS

Collection and Generics: Collection Framework Overview – Collection Interfaces – Collection Classes – Generics Concept - Generic Class – Bounded Types –Generic Interfaces.

File Handling: File Class – File Handling Basics – Byte Streams and Character Streams – FileInputStream and FileOutputStream – FileReader and FileWriter – BufferedReader and BufferedWriter – Reading and Writing Text Files – Introduction to NIO Files and Paths

Lambda Expression and Streams: Introduction to Lambda expression – Block Lambda Expressions – Functional Interfaces–Introduction to streams – Stream Creation – Intermediate and terminal operations.

Lab Exercises:

5. Implement the concept of collection framework and generics.
6. Implement the concept of lambda expressions and stream operations.

Unit-4 **Teaching Hours: 15**

SWING and JDBC

Swing Package: Origin of swing – MVC architecture- Components and Containers – Simple Swing Application–Event handling- JButton – JList – JTable – JcheckBox – JradioButton – JTextField – JComboBox- JScrollPane –JtoggleButton – JLabel – JTabbedPane – JMenuBar, JMenu – JmenuItem

JDBC: Database Programming – JDBC Drivers – Database Connectivity – Creating a Statements- Executing a queries – Processing ResultSet – PreparedStatements

Lab Exercises:

7. Develop GUI applications using Swing.
8. Implement database connectivity using JDBC.

Unit-5 **Teaching Hours: 15**

JAVA SERVLETS and JSP

Java Servlets: Servlets Basics – Servlet Life Cycle – Servlet API – GenericServlet and HttpServlet –HttpServletRequest – HttpServeltResponse– Handling Client Requests and Responses.

JSP : JSP development model – Component of JSP page – Page directive – Action– JSP Actions – Scriptlet, Expression and Declaration –

Implicit Objects – JSP with Servlets

Lab Exercises:

9. Develop dynamic web applications using Servlets
10. Develop JSP program illustrating JSP components and development model.

Text Books and Reference Books

1. Schildt Herbert, Java :The Complete Reference, Tata McGraw- Hill, 13th Edition,2024
2. The complete reference JSP 2.0, Tata McGraw- Hill, 2nd Edition, Phil Hanna,2017
3. Core Servlets and JavaServer Pages, Marty Hall and Larry Brown, Prentice Hall,2000

Essential Reading / Recommended Reading

1. Java ™ A Beginner's Guide, Herbert Schildt, McGraw-Hill Education, 10th Edition, 2024
2. Core Java: Fundamentals, Volume 1, Horstmann Cay, 12th Edition, 2021

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	L	3	2	1	1	1	1	1	1
CO2	R	3	3	3	1	1	1	1	1
CO3	N	2	2	2	3	1	2	1	1
CO4	N	2	3	3	2	1	1	1	1
CO5	G	2	2	2	3	1	1	1	1

Evaluation Pattern: CIA - 50% ESE - 50%

BCA209-3 – PYTHON PROGRAMMING

Course Name:	**PYTHON PROGRAMMING**	**Code:**	**BCA209-3**
Total Hours:	**75**	**L+P+T:**	**3+2+0**
Max Marks:	**100**	**Credits:**	**4**
Course Type:	**Major Core**	**Course Category:**	**Theory and Practical**

Course Description

This course provides a structured learning path from Python fundamentals to advanced application development. Students will begin with core programming concepts, data structures through object-oriented programming and automation, advance to GUI development, web applications and data analysis. The course emphasizes hands-on learning and real-world problem solving using modern Python libraries and frameworks.

Course Objectives

This course aims to provide a strong foundation in Python programming fundamentals and gradually introduce advanced concepts required for real-world application development. The course emphasizes problem solving, object-oriented programming, GUI and web application development, and data analysis using modern Python libraries and frameworks.

Course Outcomes

After completing this course , Students will be able to:

CO1: Understand Python fundamentals, data structures, and control structures to solve computational problems.
CO2: Design modular and object-oriented Python programs using industry-standard practices.
CO3: Develop GUI-based and web-based applications using Python frameworks.
CO4: Analyse file handling, automation, and data processing tasks using Python libraries.
CO5: Evaluate and implement data using NumPy, Pandas, and visualization tools.

Unit-1 **Teaching Hours: 15**

DATA STRUCTURES AND CONTROL STRUCTURES

Python Basics: Introduction to Python, installation, and setup, **Variables and Constants**: Naming conventions and assignments, **Data Structures**: Integers, floats, strings, booleans, Lists, Tuples, Dictionaries, and Sets, Basic Problem Solving Using Data Structures, **Control Structures**: Conditional statements (if-else), loops (for, while), **Functions**: Defining and calling functions, parameters, and return values

Lab Exercises:

1.Develop a Python application to implement basic data types and operators.

2.Apply decision-making and looping concepts to create a robust login option for your domain-specific app

Unit-2 **Teaching Hours: 15**

OOPS CONCEPTS AND REGULAR EXPRESSIONS

Classes and Objects: Defining classes, creating objects, instance variables, **Methods**: Instance methods, class methods, and static methods, **Inheritance**: Inheriting from parent classes, method overriding, **Polymorphism and Encapsulation**: Polymorphic behavior, private and protected members, **Abstraction**: Abstract classes and methods. **Regular Expressions:** Regex Syntax and Basic Patterns, Regex Operations in Python, Advanced Regex Techniques.

Lab Exercises:

3. Create a Python class for a domain based system.

4. Develop a Python application for a domain based system.

Unit-3 **Teaching Hours: 15**

GUI AND WEB PROGRAMMING

Introduction to Tkinter: GUI components, windows, and event handling, **Tkinter Widgets**: Buttons, labels, text boxes, menus, list boxes, **Layout Management**: Pack, grid, and place methods

Building a Simple GUI: Creating a basic interactive GUI application,

Streamlit Introduction: Basic Streamlit setup, creating simple apps, and displaying text/output on a web page **Streamlit for GUI**: Building a GUI-like app with Streamlit for displaying data, charts, and interactive components - deploying Streamlit applications online. **Introduction to Flask/Django**: Overview of Flask/Django, routing, and creating a simple web application

Lab Exercises:

5. Create a calculator application using Tkinter. Include basic operations such as addition, subtraction, multiplication, and division, with an interface that displays the results.

6. Design a Streamlit application that takes user input for name and age and then displays a personalized greeting message along with a simple data visualization (e.g., age vs. year of birth).

Unit-4 **Teaching Hours: 15**

FILE HANDLING AND DATA OPERATIONS

File Handling Basics: Reading and writing text files, binary files, **Advanced File Handling**: Parsing CSV, JSON, and XML files, **Error Handling**: Exception handling in file operations, **Data Handling with Pandas**: Introduction to data frames, reading and writing CSV files, data selection and manipulation.

Lab Exercises:

7.Develop a program that reads a text file containing information for e.g. names and ages of people. The program should calculate the average age and display the names of people above the average age.

8.Create a Python application to manage contacts. Allow the user to store and retrieve contact details (name, phone number, email) in a CSV file, implementing basic CRUD operations (Create, Read, Update, Delete).

Unit-5 **Teaching Hours: 15**

DATA ANALYSIS WITH NUMPY AND PANDAS

Introduction to NumPy: Arrays, matrix operations, array slicing and indexing, Data Aggregations with NumPy: Summing, mean, median, and standard deviation, Pandas Basics: Creating data frames, data selection, and manipulation, Handling Missing Data: Filling, dropping, and imputing missing values in datasets, Visualization: Plotting with Matplotlib and Seaborn,

Lab Exercises/ Capstone Project:

9. Perform statistical analysis using Numpy
10. Develop a Data-Driven Web Application for Real-World Problem Solving using Pandas

Text Books And Reference Books

1. Wesely J.Chun,Core Python Application Programming ,Prentice Hall,third edition 2015.
2. T.R.Padmanabhan, Programming with Python,Springer Publications,2016

Essential Reading / Recommended Reading

1. Zhang.Y,An Introduction to Python and Computer Programming, Springer Publications,2016

2. Sujay Ragavendra, Beginner's Guide to Streamlit with Python: Build Web-Based Data and Machine Learning Applications, Apress, 2022

3. Web Programming: Desktop Management, Aferganatel, PHI, 2004.

4. Web Technology, Rajkamal, Tata McGraw-Hill, 2001.

Web Resources:

1.https://docs.djangoproject.com/en/5.1/

2.Streamlit YouTube Channel

3.https://www.django-rest-framework.org/

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	G	3	2	3	3	2	1	1	1
CO2	G	2	3	2	2	2	1	1	1
CO3	G	2	2	2	2	2	1	1	1
CO4	G	3	3	3	3	2	1	1	1
CO5	G	3	3	3	3	2	1	1	1

Evaluation Pattern: CIA - 50% ESE - 50%

BCA210-3 - COMPUTER NETWORKS

Course Name:	**COMPUTER NETWORKS**	**Code:**	**BCA210-3**
Total Hours:	**60**	**L+P+T:**	**4+0+0**
Max Marks:	**100**	**Credits:**	**4**
Course Type:	**Major Core**	**Course Category:**	**Theory (LCA)**

Course Description

This course provides a comprehensive introduction to the fundamental concepts and principles of computer networks. Students will explore the layered architecture of modern networking systems, delving into the functionalities and services provided at each layer. The course covers key networking protocols, algorithms, and technologies, enabling students to understand how data is transmitted, routed, and managed across interconnected systems. By the end of the course, students will have a solid foundation in network operations, including protocol design, communication strategies, and the principles behind reliable and efficient data exchange.

Course Objectives

The goal of this course is to introduce the basics of computer networks. Students will learn their fundamental layered structure, understand commonly offered layered services, and examine protocols and algorithms used to operate the network.

Course Outcomes

CO1: Describe various network types, internal components, and their functionalities.

CO2: Demonstrate fundamental data communication concepts and their significance in networking.

CO3: Analyze network communication processes to diagnose issues and optimize performance.

CO4: Apply networking principles to design and implement solutions in real-world scenarios.

CO5: Exhibit network protocols and their role in efficient data communication.

Unit I **Teaching Hours: 11**

DATA COMMUNICATIONS COMPONENTS

Data communication components-Data flow – network models – categories of networks –the interconnection of networks – inter network Protocols and standards: protocols-standards-standards organizations- internet standards tasks – OSI model – layers in the OSI model – TCP/IP protocol suite.

Self Learning: Network Topology and their various Types; Types of Network: LAN, MAN, WAN; Server Based LANs & Peer-to-Peer LANs.

Unit II **Teaching Hours: 11**

PHYSICAL LAYER

Data and signals-transmission impairments - Digital Transmission: Digital to digital conversion: Line coding – line coding schemes (Unipolar, Polar, Bipolar) – block coding 4B/5B- Analog to digital conversion – PCM - transmission modes: serial transmission – parallel transmission. Analog Transmission: Digital to Analog conversion: FSK-ASK-PSK Analog to Analog conversion. Multiplexing: Frequency division multiplexing – Time division multiplexing.

Self *Learning: Introduction to Transmission Media*.

Unit III **Teaching Hours: 12**

DATA LINK LAYER

Error correction and detection: Introduction- block coding-linear block codes-cyclic codes-checksum, Hamming Code. Data link Control: protocols - simplest protocol- stop and wait for automatic repeat request--selective repeat-automatic repeat request- piggybacking.

Multiple Access: Random access-Aloha-CSMA-CSMA/CD-CSMA/CA Controlled access: reservation, polling, token passing.

Self Learning: Channelization: FDMA, TDMA, CDMA.

Unit IV **Teaching Hours:12**

TRANSPORT LAYER

Transport Layer Protocols– UDP – TCP Congestion control: Data traffic – congestion – congestion control. Application layer: Overview of Name space, DNS, FILE transfer: FTP, WWW and HTTP: Architecture – web documents – HTTP.

Self Learning: DNS using CISCO packet tracer

Unit V **Teaching Hours: 14**

WIRED LANs

Ethernet Protocol, standard Ethernet- fast Ethernet- Gigabit Ethernet. Wireless Lans: IEEE 802.11Project- architecture-MAC sublayer. Network Layer: Internet Protocol: IPv4 –IPv6 Address mapping protocols: ARP – RARP.

Routing protocols: Unicast routing protocols: distance vector routing, Link State routing Multicast Routing - Multicasting- Multicast Applications.

Self Learning: Configuring the network with multiple devices CISCO Packet Tracer.

Text Books and Reference Books

1. Behrouz A Forouzan, *Data Communications and Networking with TCP/IP Protocol Suite*, McGraw-Hill, 6th edition, 2022.
2. Behrouz A Forouzan, *Data communication and networking*, McGraw-Hill, 5th edition, 2017.
3. Andrew S Tanenbaum, *Computer Networks*, PHI publications, 5th edition, 2022.

Essential Reading / Recommended Reading

1. Mike Meyers, CompTIA Network+, 7th Edition, McGrawHill Publications, 2018.
2. James Kurose, Computer Networking: A Top-Down Approach, Pearson Publications, Global Edition, 2021.

Reference websites:

1. https://www.netacad.com/courses/packet-tracer
2. https://www.wireshark.org/download.html

3. https://www.nsnam.org/

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	G	3	1	1	2	1	2	2	2
CO2	G	3	1	1	2	1	3	2	2
CO3	G	2	3	3	3	2	1	1	2
CO4	G	1	2	3	2	3	1	2	1
CO5	G	3	1	1	2	1	1	2	2

Evaluation Pattern: CIA - 50% ESE - 50%

BCA281-3– SUMMER INTERNSHIP

Course Name:	**Summer Internship**	**Code:**	**BCA281-3**
Total Hours:	**45**	**L+P+T:**	**3+0+0**
Max Marks:	**100**	**Credits:**	**3**
Course Type:	**Internship**	**Course Category:**	**Project**

Course Description:

The Department of Computer Science offers a specialized summer internship tailored for first-year undergraduate students, spanning a duration of 4 weeks. This intensive program is designed to immerse participants in hands-on experiences encompassing computer science and application. Students will engage in practical projects and theoretical exploration under the guidance of faculty mentors, fostering a deep understanding of foundational concepts and their real-world applications. Through collaborative research, programming exercises, and data analysis tasks, interns will develop critical skills in problem-solving, algorithmic thinking, and quantitative reasoning. By the end of the internship, participants will have gained valuable insights and experience to prepare them for further academic pursuits and future careers in technology and related fields.

Course Objectives:

This course aims to provide students with practical software development experience in a real-time environment. Students will gain foundational Computer Science knowledge, develop programming proficiency, and enhance problem-solving skills through hands-on project work assigned by industry. The program also focuses on cultivating professional skills like teamwork, communication, and time management in a real-world development environment.

Course Outcomes

After completion of this course, the students will be able to

CO1: Demonstrate the ability to comprehend and apply the fundamental principles and concepts relevant to their project in Computer Applications.

CO2: Apply emerging tools and technologies in the project and ensure it aligns with the latest advancements in the field of Computer Applications.

CO3: Analyze real-time problems related to their project, utilizing various tools and techniques, and subsequently design and develop effective solutions to address those problems

CO4: Develop entrepreneurial and innovative solutions by providing sustainable and creative projects to solve real-time problems and showcasing an ability to think beyond conventional approaches

CO5: Create professional skills, critical thinking and problem-solving skills by exploring industry insights.

Unit-1 **Teaching Hours: 60**

The unit system is not adhered to for the summer internship project contents.

Text Books and Reference Books

1. "Software Engineering: A Practitioner's Approach" by Roger S. Pressman, 7th Edition, McGraw Hill Education, 2022
2. "INTRODUCTION TO ALGORITHMS, FOURTH EDITION" by Charles E. Leiserson, , Thomas H. Cormen, MIT Press, 2022
3. Head First Design Patterns: Building Extensible and Maintainable Object-Oriented Software, O'Reilly Media, 2021

Essential Reading / Recommended Reading

1. "Clean Code: A Handbook of Agile Software Craftsmanship" by Robert C. Martin, Pearson Education, 2018
2. "Introduction to the Theory of Computation" by Michael Sipser, Cengage India Private Limited, 2014
3. “The Pragmatic Programmer” by David Thomas and Andrew Hunt, Pearson India Education Services Pvt Ltd, 20th Edition, 2019

Web Resources:

1. Mozilla Developer Network (MDN) Web Docs (https://developer.mozilla.org/)
2. GitHub (https://github.com/)
3. Stack Overflow (https://stackoverflow.com/)
4. Google Developers (https://developers.google.com/)
5. The Pragmatic Bookshelf (https://pragprog.com/)

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	L,R	3	2	3	2	3	2	3	2
CO2	L,R	3	3	3	3	2	2	3	2
CO3	L,R	3	3	3	3	2	2	3	2
CO4	L,R,N,G	3	3	3	3	3	2	3	2
CO5	L,R,N,G	3	3	3	3	3	3	3	3

Evaluation Pattern: CIA - 50% ESE - 50%

FRE181-3 FRENCH

Course Name:	**French**	**Code:**	**FRE181-3**
Total Hours:	**30**	**L+P+T:**	**2+0+0**
Max Marks:	**50**	**Credits:**	**2**
Course Type:	**Theory**	**Course Category:**	**AEC**

Course learning Outcomes

CO1: To develop basic and communication skills sharpen oral and written skills: By the end of the course, students will be able to engage in basic conversations in French, demonstrating comprehension and correct pronunciation, and write short paragraphs with minimal grammatical errors.

CO2: To enhance knowledge on French culture: By the end of the course, students will be able to identify and describe at least five key aspects of French culture, such as traditions, customs, and social norms, in both oral and written formats.

CO3: To enrich the learner's vocabulary: By the end of the course, students will be able to accurately use at least 200 new French vocabulary words in context, both in speaking and writing.

CO4: To enable learners to engage in and discuss simple topics with ease: By the end of the course, students will be able to participate in simple discussions on everyday topics (e.g., family, hobbies, travel) with fluency and confidence, using appropriate sentence structures.

U1: Dossier 0 Nous découvrons le français **Teaching Hours: (4)**

U1-T1. Introduction Establish rapport

Description

Grammar | Activity | Verbe s'appeler -Les articles définis -Les pronoms sujets - Les nombres | Simple communicative wordsand phrases to develop interest. |

U2: Dossier 1 Nous apprenons le français pour..

Teaching Hours: (9)

U2-T1. Saluer -Demander et donner les informations - Donner des informations personnelles -Présenter et identifier une personne - Se presenter

Description

-Tu ou Vous -Les articles indéfinis -Les mots interrogatifs -Verbe Être -C'est ou il est -Adjectif interrogatif -Verbe avoir -Les adjectifs possessifs (1) | -Speech acts – talk about oneself. -Final task: create one's profile in a video mode. (oral communication skills, global) |

U3: Dossier 2 Nous faisons connaissance.. Teaching Hours: (9)

U3-T1. Nommer des pays et des villes - situer un lieu - Faire connaissance - Parler d'un type d'hébergement - Échanger des informations sur un hébergement

Description

- Les prépositions (noms de pays et de villes) - les prépositions de lieu et l'article contracté - négation - verbes aller, venir, Prendre - les adjectifs démonstratifs | Speech act – Asking closed questions. Final task: to create a city portrait(integrating global skills) |

U4: Dossier 3 Nous parlons la même langue Teaching Hours: (8)

U4-T1. - Parler de la famille - Décrire et caractériser un/des personnes - Parler de nos goûts - Parler de nous - Parler de nos activités - Expliquer un problèmes de santé

Description

-Les adjectifs possessifs (2) -Les adjectifs qualificatifs (M/F/Sin/Pl) -Aimer+ Adorer+ Détester +Nom/Verbe -Les verbes « er » -Le verbe faire+ du/de la / de l'/ des -Les pronoms toniques -Avoir mal à + les parties du corps | Tache finale : Introduce oneself to the class in French.(global ,communicative and speaking skills)

SEMESTER 4

BBA220-4 FINANCIAL ACCOUNTING FOR COMPUTER APPLICATION

Course Name:	**FINANCIAL ACCOUNTING FOR COMPUTER APPLICATION**	**Code:**	**BBA220-4**
Total Hours:	**45**	**L+P+T:**	**3+0+0**
Max Marks:	**100**	**Credits:**	**3**
Course Type:	**Major Core**	**Course Category:**	**Theory**

Course learning Outcomes

CO1: To understand the fundamentals of accounting, bookkeeping, principles, cycle, equation, and double-entry.

CO2: To demonstrate proficiency in recording business transactions by journalizing entries, posting to the ledger, and preparing a trial balance, using appropriate accounting principles

CO3: To analyze different types of cash books and transactions involving cash and trade discounts, solving problems related to the three-column cash book.

CO4: To prepare comprehensive final accounts, including adjustments, to evaluate a company's financial position and performance.

CO5: To apply Tally Prime and MS Excel functionalities to record transactions, and generate financial statements for effective computerized accounting in business scenarios.

Unit 1: Introduction to Accounting **Teaching Hours: 5**

Accounting Meaning, Need for accounting- Users of Accounting Information- Meaning of Bookkeeping- Accounting principles- Accounting Cycle-Phases of Accounting- Accounting Equation Double Entry Meaning- Nature and Principles of Double.

Unit 2: Accounting Process – Journal, Ledger and Trial Balance **Teaching Hours: 12**

Journal Entry Meaning and Need- Steps in Journalizing- Problems on Journal Entry-Subsidiary Books Meaning- Classification and Advantages. Ledger Meaning-Difference between Journal and Ledger-Steps in posting- Problems on Ledger Trial Balance Meaning and objectives of preparing Trial Balance-Comprehensive problems on Journal- Ledger and Trial Balance.

Unit 3: Cash Book **Teaching Hours: 7**

Cash Book: Meaning, objectives and classifications of cash book- Difference between cash discount and trade discount- Problems on Three column cash book.

Unit 4: Final Accounts **Teaching Hours: 14**

Final Accounts Meaning- Need for preparation of Final Accounts- Components of Final Accounts-Problems with adjustments

Unit 5: Computerised Accounting using Tally and MS Excel **Teaching Hours: 7**

Introduction to Computerised Accounting: Overview and Features of Tally Prime, Company Creation and Configuration in Tally, Creation of Groups and Ledgers, Recording of Accounting Vouchers (Payment, Receipt, Contra, Journal, Sales & Purchase), GST and Inventory Management in Tally, Generation of Trial Balance, Profit & Loss Account, and Balance Sheet in Tally.

Introduction to MS Excel for Financial Accounting, Recording Journal Entries in Excel, Preparation of Ledger and Trial Balance using Excel Functions, Preparation of Final Accounts using Excel, Advantages and Practical Applications of Computerised Accounting

Essential Reading

1. Fundamentals of Accounting, Universal Publications, B S Raman, 2020
2.Accounting for Managers, Himalaya Publishing House, R Venkata Ramanand Srinivas, 2020
3. S.N. Maheshwari& S.K. Gupta “Introduction to Accounting” **2020**

BCA211-4 MOBILE APPLICATIONS DEVELOPMENT

Course Name:	**Mobile Applications Development**	**Code:**	**BCA211 -4**
Total Hours:	**75**	**L+P+T:**	**3+2+0**
Max Marks:	**100**	**Credits:**	**4**
Course Type:	**Major Core**	**Course Category:**	**Theory and Practical (LCA)**

Course Description

This course provides an introduction to Android application development and the concepts involved in building mobile applications. Students will learn the fundamentals of developing, running, and debugging Android applications using modern development tools and programming techniques. The course introduces application structure, navigation, data handling, and interaction with device features and online services. Hands-on learning reinforce the theoretical concepts covered in the course.

Course Objectives

Students will be able to set up an environment for Android application development, design and develop mobile applications, manage application flow and data, integrate cloud and web services, test and debug applications, and prepare applications for deployment.

Course Outcomes

CO1: Understand the evolution, features, and versions of Android operating system.

CO2: Apply object-oriented concepts and implement Activities in Android applications.

CO3: Implement user interfaces to manage application flow and navigation.

CO4: Implement data management techniques to store and retrieve application data.

CO5: Develop Android applications that consume web services and utilize device features

Unit-1 **Teaching Hours: 15**

INTRODUCTION TO ANDROID

Introduction to Android: History and different Versions of Android, Basic Layouts and Views, Setup Android Studio, Simple Android App, Deploying the app: Running and Debugging app in Android Emulator.

Lab Exercises:

1. Implementation of different Layout and Views
2. Implement a program to perform User Interaction

Unit-2 **Teaching Hours: 15**

KOTLIN FUNDAMENTALS AND ACTIVITY

Kotlin Fundamentals: Introduction to Kotlin: Kotlin Basics – variables – Functions. Classes and Objects – Inheritance, Activity: Introduction to Activity - Activity Lifecycle – Logging, Multiple activities.

Lab Exercises:

3. Implement Functions.
4. Implement Activity Lifecycle

Unit-3 **Teaching Hours: 15**

INTENTS AND FRAGMENT

Intents, Menu, Navigation Component: back button and hierarchical structures, Fragments: Introduction - Lifecycle- Task and Back Stack, Android App Architecture

Lab Exercises:

5. Implement Intents and Menus
6. Implement Fragment Lifecycle

Unit - 4 **Teaching Hours: 15**

DATA MANAGEMENT

Data storage with Shared Preferences, SQLite, and CRUD operations. Data Binding – Live Data-Transform Live Data, Realtime Database for data management.

Lab Exercises:

7. Implement Shared Preference
8. Implement Database connectivity and CRUD operations

Unit-5 **Teaching Hours: 15**

WEB SERVICES AND DEVICE FEATURES

Networking basics: HTTP methods, JSON parsing, REST API concept, Consuming web services in Android, Runtime permissions in Android, Location services: GPS and network provider, Google Maps integration and displaying user location.

Lab Exercises:

9. Develop an application to consume REST API and display data.
10. Develop an application to display user location using Google Maps.

Text & Reference Books

[1] How to Build Android Apps with Kotlin: A Practical Guide to Developing, Testing and Publishing Your First Android Apps. Forrester, A., Boudjnah, E., Dumbravan, A., Tigcal, J. United Kingdom: Packt Publishing 2023.

[2] Nagy, Robert. Simplifying Application Development with Kotlin Multiplatform Mobile: Write Robust Native Applications for IOS and Android Efficiently. United Kingdom, Packt Publishing, 2022.

Essential Reading

[1] Griffiths, Dawn, and Griffiths, David. Head First Android Development. United States, O'Reilly Media, 2021.

[2] Kotlin for Android App Development.Sommerhoff, P. United Kingdom: Pearson Education 2018.

Additional Information (Web Resources):

1. https://www.coursera.org/learn/mastering-android-app-development-bits/home/info

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	L,N	3	2	2	2	1	2	1	1
CO2	L,G	3	2	3	2	1	2	1	-
CO3	R,N	2	2	3	2	1	1	2	-
CO4	N,G	2	3	3	2	1	1	-	1
CO5	R,G	2	3	3	3	2	2	1	1

Evaluation Pattern: CIA - 50% ESE - 50%

BCA363-4 – VISUAL PROGRAMMING

Course Name:	**Visual Programming**	**Code:**	**BCA363-4**
Total Hours:	**45**	**L+P+T:**	**3+0+0**
Max Marks:	**100**	**Credits:**	**3**
Course Type:	**SEC**	**Course Category:**	**Theory (LCA)**

Course Description

This course introduces students to the development of Windows and web-based applications using visual programming concepts and modern development tools. It focuses on designing user interfaces with standard and advanced controls, implementing file handling and exception management, and integrating applications with databases using ADO.NET. The course also provides exposure to connected and disconnected data access architectures, data binding techniques, and the migration of Windows applications to web platforms using ASP.NET. Emphasis is placed on understanding application connectivity, validation, and secure data handling in a practical and industry-relevant context.

Course Objectives

The objective of this course is to equip students with foundational knowledge of visual programming and application development using Windows Forms and web technologies. The course aims to develop the ability to design interactive user interfaces, implement database connectivity using ADO.NET with connected and disconnected models, apply data binding techniques with application controls, and understand validation, exception handling, and basic security concepts in web applications. By the end of the course, students will be prepared to build data-driven desktop and web applications with a clear understanding of modern application architecture.

Course Outcomes

After completing this course, students will be able to:

- **CO1:** Understand visual programming concepts and Windows Forms architecture using standard controls.

- **CO2:** Apply advanced controls, file handling, and exception handling concepts in Windows applications.
- **CO3:** Analyze ADO.NET architecture and implement connected and disconnected database connectivity models.
- **CO4:** Differentiate data binding techniques to integrate databases with Windows application controls.
- **CO5:** Implement web application connectivity, validation techniques, and basic security practices.

Unit-1 **Teaching Hours: 9**

WINDOWS FORMS AND VISUAL PROGRAMMING

Visual programming concepts – Event-driven programming – Windows Forms architecture – Visual Studio environment – Form design principles – Common controls: Label, TextBox, Button, ComboBox, ListBox – Container controls: Panel, GroupBox, TabControl – Basic form navigation.

Unit-2 **Teaching Hours: 9**

ADVANCED CONTROLS, MENUS AND FILE HANDLING

Dialog controls (Open, Save, Folder) – MenuStrip and ContextMenuStrip – Menu design basics – Text and binary file handling concepts – Streams overview – Exception handling basics – Try, catch, finally – Handling common runtime errors.

Unit-3 **Teaching Hours: 9**

ADO.NET ARCHITECTURE & DATABASE CONNECTIVITY

ADO.NET – ADO.NET architecture and components – Data providers – Connection, Command, DataReader, DataAdapter objects – Connected architecture – Disconnected architecture – Dataset and DataTable concepts – Basic CRUD operations through Windows applications.

Unit-4 **Teaching Hours: 9**

DATA BINDING AND APPLICATION CONTROLS

Data binding concepts – Simple and complex data binding – Binding Windows controls with databases – Grid controls overview – Using DataGridView – Introduction to stored procedures – Advantages of stored procedures – Performance and security considerations (conceptual).

Unit-5 **Teaching Hours: 9**

WEB APPLICATION AND VALIDATION

Desktop vs Web applications – ASP.NET architecture – Web forms and controls – Database connectivity in web applications – Connected and disconnected models in web context – Validation controls – Server-side exception handling – Input validation and basic security practices.

Text & Reference Books

[1] Programming in C#, E. Balagurusamy, McGraw-Hill, 2016

[2] ADO.NET: The Complete Reference, Michael Otey and Denielle Otey, McGraw-Hill, 2016

[3] Beginning ASP.NET, Matthew MacDonald, Apress (Berkeley, CA), 2012

Essential Reading

[1] C# 9 and .NET 5 – Modern Cross-Platform Development by Mark J. Price, 2020

[2] Windows Forms Programming in C# By Chris Sells, 2003

[3] Visual Basic .NET for Beginners By Peter Wright, 2025

[4] Secure Coding in C# and .NET By Mark G. Graff, Kenneth R. van Wyk, 2003

Additional Information (Web Resources):

1. https://learn.microsoft.com/dotnet
2. https://learn.microsoft.com/
3. https://learn.microsoft.com/en-us/dotnet/framework/data/adonet/

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	G	3	2	2	2	3	2	1	1
CO2	G	2	1	3	1	3	2	1	2
CO3	G	2	2	3	2	3	2	2	1
CO4	G	3	2	2	2	1	2	2	1
CO5	G	3	2	2	2	1	2	2	1

Evaluation Pattern: CIA - 50% ESE - 50%

BCA313A-4 – INTERNET OF THINGS

Course Name:	**Internet of Things**	**Code:**	**BCA313A-4**
Total Hours:	**75**	**L+P+T:**	**3+2+0**
Max Marks:	**100**	**Credits:**	**4**
Course Type:	**Core Elective**	**Course Category:**	**Theory and Practical**

Course Description

This course offers a foundational understanding of the Internet of Things (IoT), covering its core concepts, principles, and significance in modern technology. Students will explore various IoT protocols, delve into the technical and practical challenges associated with IoT implementations, and examine real-world applications across different industries. Through this course, students will gain insights into how IoT systems are designed, deployed, and managed, preparing them to address the complexities of this rapidly evolving field.

Course Objectives

This course is an introduction into the basic concepts and principles of the Internet of Things. It outlines the different protocols for the Internet of Things. The challenges and applications of IoT implementations are also discussed in detail in this course.

Course Outcomes

CO1: Understand the definition and significance of the Internet of Things.

CO2: Demonstrate the use of different protocols and architecture of the Internet of Things.

CO3: Apply the knowledge to interface various sensors with IoT development board.

CO4: Design and implement IoT system for real-time applications.

CO5: Identify and address security and connectivity challenges in IoT systems.

Unit-1 **Teaching Hours: 15**

Introduction to The Internet of Things

Evolution of IoT – IoT Definition – Elements of an IoT ecosystem – Development boards – Arduino – Raspberry Pi – BeagleBone – ESP8266 – Interfacing of sensors with development boards – Operating systems for IoT.

Lab Exercises:

1. Arduino UNO / Raspberry Pi board and components using Tinkercad
2. Blinking LED sketch with Arduino UNO / Raspberry Pi using Tinkercad

Unit-2 **Teaching Hours: 15**

Sensors

Introduction – Sensors – Roles of Sensors and Actuators – Types of Sensors – Working of Sensors – Sensor Characteristics – Sensorial Deviations – Sensing Types – Scalar sensing – Multimedia sensing – Hybrid sensing – Virtual sensing – Sensing Considerations – Wearable sensors and their applications.

Lab Exercises:

3. Interfacing of Temperature Sensor with Arduino / Raspberry Pi and displaying data on Serial Monitor
4. Room Temperature Monitoring System using Arduino / Simulation of 4-Way Traffic Light with Arduino UNO/Raspberry pi.

Unit-3 **Teaching Hours: 15**

Actuators and Control Systems

Actuators – Actuator Types – Mechanical actuators – Hydraulic actuators – Thermal or Magnetic actuators – Electric actuators – Actuator Characteristics.

Lab Exercises:

5. Simulation of Four-Way Traffic Light Control System using Arduino UNO / Raspberry Pi
6. DC Motor Control and Speed Regulation using Arduino

Unit-4 **Teaching Hours: 15**

IoT Architecture and Communication Protocols

Layered Architecture for IoT – Protocol Architecture of IoT – Constrained nodes – Constrained networks – Types of constrained devices – Low power and lossy networks – Infrastructure protocols – Internet Protocol Version 6 (IPv6) – Routing Protocol for Low-Power and Lossy Networks (RPL).

Lab Exercises:

7. Interfacing Motion Sensor for Human Movement Detection
8. Obstacle Detection using IR Sensor
9. Interfacing Gas Sensor as an IoT Edge Device
10. Spinning a DC Motor and Motor Speed Control as an IoT Actuation Example

Unit-5 **Teaching Hours: 15**

IoT Security and Applications

IoT Security and System-Level Challenges: IoT Security Threats – IoT Security Requirements – Privacy in IoT Networks – Reliability Challenges – Interoperability of Devices – Reliability Issues due to Energy Constraints.

IoT Communication Technologies and Applications: IEEE 802.15.4 – Near Field Communication (NFC) – RFID – ZigBee – Bluetooth Low Energy (BLE) – Long Term Evolution–Advanced – Z-Wave – Components of Z-Wave Network – LoRa – NB-IoT – Wi-Fi.

Lab Exercises:

11. Study of IoT Security Issues and Reliability Challenges
12. Study of IoT Communication Technologies and Their Applications

Text Books and Reference Books

[1] Introduction to IoT, Sudip Misra, Anandarup Mukherjee, Arijit Roy (Cambridge University Press) 2021

[2] Internet of Things: Principles and Paradigms, R. Buyya and A.K. Dastjerdi (Elsevier) 2016.

[3] Internet of Things: A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, 2023 (Universities Press)

Essential Reading / Recommended Reading

1. Internet of Things-Architectures, Protocols and Standards, Simone Cirani, Gianluigi Ferrari, Marco Picone, Luca Veltri, Wiley,2018
2. Internet of Things- Architecture and Design Principles, Raj Kamal, McGraw Hill Education,2017

Web Resources:

1. https://www.tinkercad.com/
2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	L,R	3	1	1	2	1	2	1	2
CO2	L,R	3	2	2	3	2	2	1	2
CO3	L,R,N	2	3	3	3	2	2	1	2
CO4	L,R,N,G	2	3	3	3	3	2	2	2
CO5	L,R,N,G	2	2	3	2	1	2	1	**1**

Evaluation Pattern: CIA - 50% ESE - 50%

BCA313B-4 – AUGMENTED REALITY AND VIRTUAL REALITY

Course Name:	**Augmented Reality And Virtual Reality**	**Code:**	**BCA313B-4**
Total Hours:	**75**	**L+P+T:**	**3+2+0**
Max Marks:	**100**	**Credits:**	**4**
Course Type:	**Core Elective**	**Course Category:**	**Theory and Practical**

Course Description

This course introduces students to the fundamentals of Augmented Reality and Virtual Reality with a strong foundation in 3D graphics and modeling. The course covers core concepts of immersive technologies, including 3D environment creation, virtual interaction design, AR/VR hardware and software platforms, and basic scripting for interactive applications.Students will gain hands-on experience using industry-standard tools such as Unity, Blender/3ds Max, ARKit, and ARCore to design and develop simple AR/VR applications, and apply these skills to conceptualize and develop domain-based projects that address SDG-aligned real-world problems through innovative AR/VR mobile/web solutions with unique features, structured functional modules, and a comprehensive project report.

Course Objectives

The objective of this course is to provide students with a strong foundation in 3D graphics and immersive technologies required for developing Augmented Reality and Virtual Reality applications. The course aims to equip students with knowledge of 3D modeling, virtual environment design, VR and AR system principles, hardware and software platforms, and basic scripting techniques. Through hands-on laboratory sessions and a mini project, students will develop practical skills to design and implement interactive AR/VR applications, gain awareness of current trends and future opportunities in immersive technologies across various domains, and frame domain-based projects that identify SDG-aligned real-world problems and develop innovative AR/VR mobile/web solutions with unique novelty, functional modules, and a comprehensive project report.

Course Outcomes

CO1: Understand 3D concepts to build 3D scenes, working principles, and explore applications of Virtual Reality (VR).

CO2: Design 3D models, integrate them into a game engine, and compare the fundamentals of Augmented Reality.

CO3: Apply AR/VR hardware, software development tools, and SDKs using Unity.

CO4: Implement interactive AR/VR applications by implementing 3D modeling, spatial mapping and user interaction techniques in Unity.

CO5: Analyze emerging trends, and design an innovative SDG-aligned AR/VR solution

Unit-1 **Teaching Hours: 15**

3D FOUNDATIONS & AUGMENTED REALITY (AR)

3D Concepts: Vertices, edges, faces, Meshes, textures, materials - Transformations: Translation, Rotation, Scaling - Cameras, lighting, and rendering pipeline, lighting, shadows – Camera setup and basic animation techniques Scene design principles

Augmented Reality (AR) and its evolution - AR systems working mechanism: Sensors, cameras, and real-world overlays - AR Development Platforms and Tools (Unity, Unreal Engine, WebAR, ARKit, ARCore) - Designing AR Experiences: Interaction principles and UI/UX considerations

Lab Exercises:

1. Creating 3D objects with appropriate materials, lighting, and shadows.
2. Creating a 3D scene in Unity with player movement and camera control.

Unit-2 **Teaching Hours: 15**

VIRTUAL REALITY FOUNDATIONS

Virtual Reality - AR VS VR: Key differences and use cases - Core Components of VR Systems – Hardware – Software - Key Features of VR: Immersion, interactivity, presence - Creating a Virtual Environment: 3D spaces, objects, and interactions - Benefits & Challenges of VR: Applications in gaming, education, healthcare, and industry - Challenges & Future Trends

Lab Exercises:

3. Storyboard creation using VR scenes.
4. Implementing basic object interactions: grabbing, rotating, and scaling.

Unit-3 **Teaching Hours: 15**

AR & VR HARDWARE AND SOFTWARE SYSTEMS

Overview of AR and VR hardware devices- Comparison of AR and VR Hardware - Sensors and tracking systems- Input devices: controllers, hand tracking, and gesture-based interfaces AR & VR Software Development - Overview of AR & VR SDKs - Setting Up a Development Environment - Installing and configuring AR Foundation in Unity

Lab Exercises:

5. Story board creation and optimizing with AR smart glasses/ VR headsets
6. Story board creation and visualization using controllers

Unit-4 **Teaching Hours: 15**

SPATIAL MAPPING AND INTERACTION DESIGN

Spatial Mapping and Surface Detection in AR - How AR Systems Detect Surfaces - Spatial Mapping in AR - Surface Detection Algorithms - Designing User Interactions with Virtual Objects - Basic Interaction Techniques - Implementing Interactions in a Real-World Scenario.

Lab Exercises:

7. Setting up an AR project in Unity with surface detection (plane tracking).
8. Placing and interacting with virtual objects in the real-world environment.

Unit-5 **Teaching Hours: 15**

Scripting & Future Trends

Scripting for AR/VR applications – C# scripting in Unity – Event-driven interaction models — Performance optimization in AR/VR – VRML (Virtual Reality Modeling Language) – Emerging trends in immersive technologies: Metaverse, Artificial Intelligence in AR/VR

Mini Project: Frame a project by selecting a domain (education, healthcare, environment, etc.) and identifying an SDG-aligned Real-World Problem; design an innovative AR/VR mobile/web solution

with Novelty and 3–4 functional modules demonstrating usability and emerging technology integration, and prepare a project report.

Text & Reference Books

[1] Arijan Belec, Blender 3D Incredible Models, Packt Publishing Limited, 2022

[2] G.C. Burdea & P. Coiffet, Virtual Reality Technology: Website Associated W/Book (IEEE Press), 2024

[3] D. Schmalstieg and T. Höllerer. Augmented Reality: Principles and Practice. Addison-Wesley, Boston, 2016

[4] Steven M. LaValle. Virtual Reality. Cambridge University Press, 2017

Essential Reading

[1] Samuel Greengard, Virtual Reality, The MIT Press, 2019

[2] Rick Parent, Computer Animation: Algorithms & Techniques, Morgan, 2012

[3] Grigore C. Burdea, Philippe Coiffet , Virtual Reality Technology, Wiley 2016

[4] Alan B. Craig, Understanding Augmented Reality, Concepts and Applications, Morgan Kaufmann, 2013.

Additional Information (Web Resources):

1.ARCore by Google - https://developers.google.com/ar

2. ARKit by Apple - https://developer.apple.com/augmented-reality/arkit/

3. Vuforia - https://developer.vuforia.com/

4. Unity3D - https://unity.com/

5. Unreal Engine - https://www.unrealengine.com/en-US

6. Oculus Developer - https://developer.oculus.com/

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	L, R, N	2	3	1	2	2	2	2	2

CO2	L, R, N	1	3	2	3	2	1	1	1
CO3	L, R, N	1	2	2	3	1	1	2	2
CO4	L, R,N, G	2	3	1	3	3	2	1	2
CO5	L, R,N, G	2	3	1	3	3	2	1	2

Evaluation Pattern: CIA - 50% ESE - 50%

BCA313C-4 – DRONE PROGRAMMING

Course Name:	**Drone Programming**	**Code:**	**BCA313C-4**
Total Hours:	**75**	**L+P+T:**	**3+2+0**
Max Marks:	**100**	**Credits:**	**4**
Course Type:	**Core Elective**	**Course Category:**	**Theory and Practical**

Course Description

This course provides an in-depth understanding of drone technologies, their architecture, and real-world applications across industries. Students will learn about drone dynamics, flight principles, propulsion systems, sensor integration, and autonomous control concepts. The course also explores regulatory aspects, ethical considerations, and recent trends in unmanned aerial systems (UAS). It prepares learners to analyze, evaluate, and innovate drone-based solutions for applications in surveillance, agriculture, logistics, and environmental monitoring.

Course Objectives

The objective of this course is to equip students with a comprehensive knowledge of drone systems and technologies, focusing on flight mechanics, sensor integration, control systems, and industrial applications. Students will also understand the regulatory framework and emerging trends shaping the future of drone ecosystems.

Course Outcomes

CO1: Understand the fundamental principles, classifications, and evolution of drones and unmanned aerial systems.

CO2: Analyze the components and working mechanisms of drone hardware including propulsion, sensors, and control systems.

CO3: Apply flight dynamics, navigation, and communication systems used in drones.

CO4: Evaluate safety standards, regulations, and ethical aspects associated with drone operations.

CO5: Discuss emerging trends and applications of drone technologies in various sectors.

Unit-1 **Teaching Hours: 15**

INTRODUCTION TO DRONE SYSTEMS

Evolution and classification of drones – Fixed-wing, rotary-wing, and hybrid UAVs - Principles of flight and aerodynamics - Terminologies: UAV, UAS, RPAS - Components overview – Frame, motors, propellers, and batteries - Applications and trends in drone technologies

Lab Exercises:

1. Program drone for take-off and landing in Developer Mode.
2. Control onboard LEDs and verify signal response via controller app.

Unit-2 **Teaching Hours: 15**

DRONE ARCHITECTURE AND COMPONENTS

Hardware structure and functionality - Sensors: Accelerometer, gyroscope, magnetometer, barometer, GPS, ToF, and LiDAR - Communication modules and telemetry systems - Flight control boards and electronic speed controllers (ESCs) - Power systems and payload integration

Lab Exercises:

3. Collect and visualize sensor data from accelerometer and gyro.
4. Activate and control onboard camera using Pluto Blocks or app interface.

Unit-3 **Teaching Hours: 15**

FLIGHT MECHANICS AND CONTROL SYSTEMS

Axes of rotation and stability mechanisms - Control surfaces, thrust, lift, and drag principles - Propulsion systems and flight controllers - Autopilot concepts and navigation algorithms - Fundamentals of drone communication and remote operation

Lab Exercises:

5. Build programs to monitor propeller speed at varying levels
6. Perform a diagnostic test on ESC and motor interconnectivity.

Unit-4 **Teaching Hours: 15**

REGULATIONS, SAFETY AND ETHICS

Airspace classifications and DGCA/FAA regulations - Licensing, registration, and permissions for drone operations - Risk assessment and safety guidelines - Ethical implications and societal impact of drones - Environmental considerations and data privacy concerns

Lab Exercises:

7. Various propeller speeds using PWM with motor signals.

8. Perform a diagnostic test on ESC and motor interconnectivity.

Unit-5 **Teaching Hours: 15**

EMERGING TRENDS AND APPLICATIONS

Drones in agriculture, mapping, logistics, and surveillance - Drone-based disaster management and emergency response - Integration with AI, IoT, and 5G - Future prospects: Swarm drones, delivery drones, and autonomous aerial vehicles - Case studies of successful drone innovations

Mini-Project: Conceptualizing and developing a drone-based solution

Case Studies: Agricultural Weed Detection, Microdrone Systems, Aerial Surveillance

Text & Reference Books

[1] Reg Austin, Unmanned Aircraft Systems: UAV Design, Development and Deployment, Wiley, 2010.

[2] Kimon P. Valavanis, Advances in Unmanned Aerial Vehicles: State of the Art and the Road to Autonomy, Springer, 2007.

[3] Richard Williams, Programming Drones with Python, Packt Publishing, 2021.

Essential Reading

[1] Cybellium Ltd., Mastering Drone Design and Programming: A Comprehensive Guide, 2023.

[2] Neeraj Kumar Singh, Porselvan Muthukrishnan, and Satyanarayana Sanpini, Industrial System Engineering for Drones, Apress, 2019.

Additional Information (Web Resources):

1. https://droneblocks.io/program/programming-a-drone
2. https://tello.oneoffcoder.com/index.html
3. https://www.dronaaviation.com/plutox/

CO – PO Mapping

	Development Needs (LRNG)	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO 1	G	3	2	2	2	2	1	1	1
CO 2	G	3	3	2	2	2	1	1	1
CO 3	G	2	3	3	2	2	2	1	1
CO 4	G	2	2	3	3	2	2	2	1
CO 5	G	2	3	3	3	3	2	3	2

Evaluation Pattern: CIA - 50% ESE - 50%

BCA381-4 DBMS PROJECT

Course Name:	**DBMS Project**	**Code:**	**BCA381-4**
Total Hours:	**60**	**L+P+T:**	**0+4+0**
Max Marks:	**100**	**Credits:**	**2**
Course Type:	**Major Core**	**Course Category:**	**Practical**

Course Description

This course provides hands-on experience in designing and implementing, and managing a relational database system based on real-world requirements. Students will transform theoretical knowledge of database modeling into practical applications by working in teams to identify a problem, model the data using Entity-Relationship diagrams, normalize the schema to avoid anomalies, implement the database using SQL, and develop a frontend interface using .NET framework to interact with the database.

Course Objectives

The course aims to enable students to bridge the gap between theoretical database concepts and practical implementation by designing, creating, and manipulating a functional database system, incorporating normalization, advanced SQL querying, and database integrity.

Course Outcomes

CO1: Analyze real-world business scenarios to identify data requirements and construct comprehensive Entity-Relationship (E-R) diagrams and relational models.

CO2: Apply normalization techniques up to BCNF to develop efficient and well-structured database schema.

CO3: Construct complex SQL queries, including multi-table joins, nested queries, and set operations, for effective data manipulation and retrieval.

CO4: Develop backend database objects such as Views, Stored Procedures, Cursors, and Triggers to automate database tasks and ensure data integrity.

CO5: Design and develop a front-end application interface using .NET that interacts with the backend database to perform CRUD (Create, Read, Update, Delete) operations.

Unit-1

The unit system is not adhered to for this project contents, but will include the following phases:

Project Phases:

Project Initiation & Requirements Analysis: Selecting the project topic, identifying stakeholders, and collecting functional/non-functional requirements.

Conceptual Design (ER Modeling): Creating Entities, Attributes, and Relationships, and drawing the Entity-Relationship Diagram (ERD).

Logical & Physical Design (Normalization): Mapping ERD to relations, normalization (up to 3NF/BCNF), and determining data types and constraints,

Implementation: Choosing of SQL platform, writing DDL/DML scripts, creating tables, and establishing the database using triggers/procedures for complex constraints and automation.

Application Front-End Development: Students are required to use the **.NET framework** to design the user interface and establish database connectivity.

Testing, Documentation & Final Presentation: Running test cases using CRUD operations, finalizing documents, and demonstrating the application.

Text & Reference Books

[1] Ramez Elmasri and Shamkant B. Navathe, *Fundamentals of Database Systems*, 7th Edition**,** Pearson, 2015.

[2] Damini Grover, *Dot Net Technology*, I.K. International Publishing House Pvt. Limited 2011.

[3] Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom, *Database Systems: The Complete Book,* 2008.

Essential Reading

[1] Connolly, T. M., Begg, C. E. Database Systems: A Practical Approach to Design, Implementation, and Management, 6th edition, Pearson, 2019.

[2] A S Modi, *Dot Net Technologies*, B R Publishers, 2022.

Additional Information (Web Resources):

[1] NPTEL (Swayam - Database Management System)

[2] GeeksforGeeks DBMS Tutorial: https://www.geeksforgeeks.org/dbms/dbms/

CO – PO Mapping

	Development Needs (LRNG)	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	L, R	1	2	3	1	2	1	1	2

CO2	L, R	2	3	2	2	2	2	1	3
CO3	L, R	2	2	2	2	2	1	2	3
CO4	L,R, N, G	2	3	2	3	2	2	2	1
CO5	L,R, N, G	1	3	3	2	2	1	2	2

Evaluation Pattern: CIA - 50% ESE - 50%

FRE181-4 FRENCH

Course Name:	**French**	**Code:**	**FRE181-4**
Total Hours:	**30**	**L+P+T:**	**2+0+0**
Max Marks:	**50**	**Credits:**	**2**
Course Type:	**AEC**	**Course Category:**	**Theory**

Course learning Outcomes

CO1: To develop basic and communication skills sharpen oral and written skills: By the end of the course, students will be able to engage in basic conversations in French, demonstrating comprehension and correct pronunciation, and write short paragraphs with minimal grammatical errors.

CO2: To enhance knowledge on French culture: By the end of the course, students will be able to identify and describe at least five key aspects of French culture, such as traditions, customs, and social norms, in both oral and written formats.

CO3: To enrich the learner's vocabulary: By the end of the course, students will be able to accurately use at least 200 new French vocabulary words in context, both in speaking and writing.

CO4: To enable learners to engage in and discuss simple topics with ease: By the end of the course, students will be able to participate in simple discussions on everyday topics (e.g., family, hobbies, travel) with fluency and confidence, using appropriate sentence structures.

U1: Dossier 4- Nous parlons de notre quotidien **Teaching Hours: (10)**

Indiquer l'heure et les horaires -Parler de nos activités et de nos habitudes quotidiennes. -Parler de notre journée de travail. -Parler de nos sorties -Inviter et accepter ou refuser une invitation

Description

Grammar | Activity | -Quelques articulateurs de temps -L'heure -Les verbes pronominaux -Verbes : Lire, écrire -L'expression de l'habitude, et de la fréquence -Verbes : Choisir, Sortir -Poser des questions L'impératif | Tache finale : Organise an outdoor activity, a day in nature. |

U2: Dossier 5 Nous nous informons en français **Teaching Hours: (10)**

. ▪ Raconter des événements passés ▪ Parler d'expériences ▪ Comprendre des informations biographiques ▪ Décrire physiquement une personne ▪ Donner des conseils.

Description

▪Le Passé Composé avec avoir ▪Le Passé Récent ▪Le Futur Proche ▪Être/ Avoir+ Adjectif ▪L'impératif Présent | Create a virtual space to describe a stranger. |

U3: Étape 6 Nous rêvons d'aller dans un pays francophone Teaching Hours: (10)

Comprendre le programme d'un séjour. ▪ Choisir une destination et une formule de voyage ▪ Caractériser une ville ▪ Décrire une hébergement ▪ Parler des saisons.

Description

▪Les adjectifs qualificatifs ▪Des structures pour parler du climat ▪Le Futur simple ▪Le pronomY | Organise a virtual tour to a Francophone country. |

Essential Reading

https://cosmopolite.hachettefle.fr/cosmopolite-1_livre-de-l-eleve_en.html

BCA281-4– SERVICE LEARNING

Course Name:	**Service Learning**	**Code:**	**BCA281-4**
Total Hours:	**30**	**L+P+T:**	**2+0+0**
Max Marks:	**100**	**Credits:**	**2**
Course Type:	**VAC**	**Course Category:**	**Theory– Community Engagement**

Course Description

Service-learning course provides an opportunity for students to integrate their computer technical knowledge and skills with community service. Students will engage in a service-learning project that applies their programming skills, algorithms and software applications, data management, and development of software applications for a real-world setting. Communication in a real-world setting. Through the project, students will develop leadership, teamwork, and problem-solving skills while making a constructive impact in their community.

Course Objectives

This course enables students to apply their computer science knowledge—such as programming, data handling, and software development—to address real needs within the community.

Course Outcomes

CO1: To understand the responsibility of a Computer Science student and to apply computer applications skills to a service-learning project.

CO2: To analyze the social needs, and issues and develop software solutions that meet the needs of the community.

CO3: To demonstrate the transformative knowledge building through community service using academic comprehension for the social issues

Unit-1 **Teaching Hours: 4**

SERVICE LEARNING OVERVIEW

Introduction to Service Learning – Need for Service Learning-Difference between Learning, Service, Social Service and Service Learning – Benefits – The four phases: Preparation, Action, Reflection and

Evaluation – Service Learning Diary - different models for integrating service-learning into the curriculum

Outcomes – Personal (Learning and Career) - Interpersonal, Society level outcomes.

Unit-2 **Teaching Hours: 6**

STUDENT PARTICIPATION IN SERVICE LEARNING

Choosing the target Society / Community - Problem identification in the chosen area – Motivation for SL – Analyze various Problem-solving strategies and propose the solution –Execution Plan - Design a service learning project-Identify community needs assessment, Reflection and Impact analysis, Report submission on the problem Identification, Solution strategies analysis and identify the best solution and the plan of execution of the proposed solution and confirming the Service Learning Project.

Unit-3 **Teaching Hours: 20**

COMMUNITY PROJECT

Implementation of chosen Service Learning Project –Identification and overcoming of Challenges – Impact analysis - Critical Reflections – Report submission – Evaluation of the progress at the end of every Five hours - assess and grade service-learning.

Project Phases:

1. **Community Partner Engagement:**
   - **Identification:** Research and identify potential community partners whose needs align with student skills and interests. Consider organizations focused on education, social services, environmental sustainability, or other areas.
   - **Needs Assessment:** Conduct meetings or interviews with the partner to understand their specific needs and challenges. Clearly define the problem the project will address.
   - **Project Scope Definition:** Collaboratively define the project scope, deliverables, and timeline with the partner. Ensure the project is realistic and achievable within the given timeframe. Document this agreement in a project proposal or memorandum of understanding.
2. **Project Design & Development:**
   - **Technical Design:** Develop a detailed technical plan, including chosen technologies, system architecture, data flow, and user interface design (if applicable).

- **Development & Implementation:** Build and test the software or system, adhering to best practices for coding, documentation, and quality assurance.
- **User Training (if applicable):** Develop training materials and provide training to the partner's staff or users on how to use the developed system.

3. **Project Deployment & Evaluation:**
   - **Deployment:** Deploy the project in the partner's environment, ensuring a smooth transition and minimal disruption.
   - **Testing & Feedback:** Gather feedback from the partner and users on the project's functionality, usability, and impact.
   - **Evaluation:** Evaluate the project's success based on the agreed-upon metrics and the partner's feedback. Document the evaluation results.
4. **Reflection & Reporting:**
   - **Individual Reflection:** Students will reflect on their learning experiences, challenges faced, and the project's impact on the community.
   - **Team Report:** The team will prepare a comprehensive report documenting the project's goals, design, development process, results, and lessons learned.
   - **Presentation:** The team will present their project to the community partner, faculty, and other students.

Essential Reading

[1] Barbara Jacoby, Jeffrey Howard, Service-Learning Essentials: Questions, Answers, and Lessons Learned, Wiley Publications, 2014.

[2] Brian A. Nejmeh, Service‑Learning in the Computer and Information Sciences: Practical Applications in Engineering Education, Wiley-IEEE Press z,2012

Additional Information (Web Resources):

1. https://epics.ieee.org/projects/
2. https://epics.engineering.asu.edu/epics-projects/
3. https://epic.uchicago.in/projects/
4. Service Learning in Computer Science https://cce.sonoma.edu/

CO – PO Mapping

	Development Needs (LRNG)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	L,R	3	2	2	2	1	2	1	1
CO2	L,R	2	2	2	2	2	2	2	1
CO3	L,R	2	2	2	2	2	2	2	1

Evaluation Pattern: CIA - 50% ESE - 50%

ANNEXURE I

Project Guidelines:

The student should choose a proper community agency and a project within the context of current societal problems and apply their programming skills, algorithms and software applications, data management, and the development of software applications to solve the same; present their outcomes in front of a panel along with the feedback of community agency and a report in the prescribed format.

Types of Service Learning Projects:

A) Software Development Projects:

- **Web Applications:** Creating websites or web apps for non-profits, community centers, or schools. Examples include:
    - A website for a local animal shelter to showcase adoptable pets.
    - A web application for a community garden to manage plots and volunteer sign-ups.
    - An online platform for a literacy program to connect tutors and students.
- **Mobile Applications:** Developing mobile apps to address community needs. Examples include:
    - A mobile app for a food bank to distribute information about food distribution events.
    - A mobile app for senior citizens to access community resources and connect with family.
    - A game-based learning app for children in an underserved school.
- **Desktop Applications:** Building desktop software for specific tasks. Examples include:
    - A database application for a historical society to manage its archives.
    - A program for a community health clinic to track patient data (with appropriate privacy safeguards).
    - A tool for a local environmental group to analyze data collected from their monitoring efforts.

B) Data Analysis and Visualization Projects:

- **Data Collection and Analysis:** Helping organizations collect, clean, and analyze data to improve their programs or services. Examples include:
    - Analyzing data for a social service agency to identify trends in client needs.
    - Analyzing environmental data for a local conservation group to track pollution levels.
    - Analyzing student performance data for a school district to identify areas for improvement.
- **Data Visualization:** Creating visualizations to communicate complex data in a clear and understandable way. Examples include:
    - Developing interactive dashboards to visualize data for a non-profit's annual report.
    - Creating maps to visualize the distribution of resources in a community.
    - Designing infographics to communicate key findings from a research study.

C) IT Support and Infrastructure Projects:

- **Network Setup and Maintenance:** Helping community organizations set up and maintain their computer networks. Examples include:
    - Setting up a wireless network for a community center.
    - Troubleshooting computer problems for a senior center.
    - Providing IT training to staff at a non-profit organization.
- **Database Management:** Designing and implementing databases for organizations to manage their data more effectively. Examples include:
    - Creating a database for a library to manage its collection.
    - Developing a database for a volunteer organization to track its volunteers.
    - Migrating an organization's data to a more efficient and secure system.

D) Educational and Training Projects:

- **Coding Workshops:** Teaching coding skills to students in underserved communities. Examples include:
    - Running after-school coding clubs for middle school students.
    - Organizing workshops for high school students to learn web development.
    - Teaching adults basic computer skills to improve their job prospects.

- **Technology Training:** Providing training to community members on how to use different software applications or technologies. Examples include:
    - Teaching senior citizens how to use email and the internet.
    - Training non-profit staff on how to use social media for outreach.
    - Developing online tutorials for community members to learn new software skills.

E) User Accessibility Projects:

- **Assistive Technology:** Developing or adapting assistive technology for individuals with disabilities. Examples include:
    - Creating software to help people with visual impairments access information online.
    - Developing mobile apps to support communication for people with autism.
    - Designing accessible websites for organizations serving people with disabilities.

E) Cybersecurity Projects:

· **Security Audits and Training:** Helping community organizations improve their cybersecurity practices.